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Clinical Management of the Adult Kidney Transplant Recipient

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CHAPTER OUTLINE

KEY POINTS

- Kidney transplantation remains the preferred modality for managing individuals with kidney failure, with superior outcomes in terms of patient survival and quality of life.
- Availability of a sufficient supply of kidneys is a key challenge worldwide, leveraging live donation, paired kidney donation, donor management strategies including ex-vivo machine-based perfusion, xenotransplantation, and allocation schemes to better serve those waiting for a kidney transplant.
- Immunosuppression is required long term for patients receiving a kidney transplant, and therapy includes both induction and maintenance treatment. Only in the past couple of years have new medications been developed particularly for antibody-mediated rejection.
- As for native renal disease, the initial evaluation of transplant allograft dysfunction should be approached systematically using an anatomic classification of prerenal, intrarenal, and postrenal causes, with important consideration regarding the timing of an event post transplant.
- Long-term graft survival has improved modestly in the past 20 years; death due to cardiovascular disease with a functioning transplant remains a key challenge, as well as graft failure due to subclinical cellular inflammation or the injury mediated by donor-specific human leukocyte antigen antibodies.

The chapter topics represent the journey of a patient with kidney failure as they traverse the landscape of the transplant process. Orienting toward the patient evaluation for transplant candidacy, this chapter covers the surgical procedure, post-transplant care including immunosuppression, common transplant complications such as infection and rejection, and long term outcomes including allograft failure and death with a functioning graft. Management strategies are discussed where relevant.

MEDICAL EVALUATION OF THE RECIPIENT BEFORE TRANSPLANTATION

The potential kidney transplant recipient should undergo a thorough evaluation to ensure that there are no contraindications to transplant surgery or long-term immunosuppression. Early referral is key to increasing the chances of preemptive kidney transplantation, which results in better posttransplant outcomes, especially in children.[1] Wait time begins at the time of dialysis initiation or at the time of wait list registration once the eGFR falls below a certain locally defined threshold (e.g., <20 mL/min/1.73 m^2 (whichever happens first). The evaluation process includes a rigorous assessment of the candidate's medical history including cardiac evaluation (particularly important given the significant burden of cardiovascular disease in individuals with kidney failure), infection, age-appropriate cancer screening, and assessment of immunologic risk (i.e., risk of allograft rejection). Identification of the cause of native kidney disease is particularly important, as it may impact peritransplant and post-transplant management and risk of recurrent disease, discussed later.

SPECIAL RECIPIENT CHARACTERISTICS

FRAILTY AND OLDER AGE

Frailty, while challenging to assess clinically, is associated with poor outcomes post transplant. One study found that frail patients (median age 54 years) were less likely to be listed for kidney transplantation. Assessment of frailty included unintentional weight loss, grip strength, walking speed, exhaustion, and activity level. Though some transplant centers may have an upper age threshold for transplant eligibility, 5-year patient survival was >70% in a cohort of recipients older than 60 years old. In a subgroup of recipients 70 years and older, 5-year survival was 78%.[2] As with all transplant recipients, careful patient selection and shared decision making are paramount. Lower immunosuppression may be advisable in the older population.

OBESITY

There is significant center-to-center variability with respect to transplantation of patients with obesity; some centers have >20% of recipients with body mass index (BMI) >35 while others have <5%.[3] Severe obesity (BMI 35 kg/m^2 or greater) has been shown to be associated with more transplant surgery–related complication, delayed graft function (DGF), and poorer allograft survival.[4,5] A 2021 meta-analysis of UNOS data that compared BMI categories of transplant recipients (median follow-up 4 years) found increased DGF in patients with BMI >30 kg/m^2 and no difference in allograft or patient outcomes between BMI >30 to 35 kg/m^2 and BMI >35 kg/m^2. There was no difference in patient survival or hospital length of stay following transplant across BMI categories. This study supports the idea of greater flexibility in BMI inclusion criteria for potential kidney transplant recipients. Further, a study of transplant recipients with BMI >30 kg/m^2 found that transplantation provides a survival benefit over remaining on the waiting list (receiving dialysis), up to a BMI of 41 kg/m^2.[6] Bariatric surgery before transplantation can be considered for severely obese patients.[7] Body surface area (BSA) mismatch between donors and recipients has been associated with poorer long-term allograft survival.[8,9] With the advent of the GLP1 receptor agonists, the management of potential recipients with an elevated BMI may change in the future.

HUMAN IMMUNODEFICIENCY VIRUS INFECTION

Though HIV infection was traditionally considered an absolute contraindication to kidney transplantation due to concerns for infections and short patient survival, improvements in HIV treatments and transplantation of patients with HIV have led to favorable outcomes.[10,11] On the basis of inclusion criteria from a 2010 study, patients should be maintained on highly active antiretroviral therapy, have a CD4$^+$ T cell count of at least 200 cells/mm^3, and undetectable plasma HIV RNA levels. Kidney transplant recipients with HIV infection appear to be at higher risk of acute rejection and thus may benefit from antithymocyte globulin (ATG) induction.[12] The HOPE Act, signed into law in 2013, called for the use of organs from HIV-positive donors for transplantation into HIV-positive candidates under approved research protocols. As of 2020, 170 such kidney transplants have been performed in the United States as a result. More recently, living donors with HIV have successfully donated to recipients with HIV.[13]

DIABETES AND KIDNEY-PANCREAS TRANSPLANTATION

Although the survival rate of patients with diabetes after kidney transplantation is lower than that of those without, the benefits of transplantation versus staying on dialysis remain.[14] A subset of potential transplant candidates may be suitable for kidney-pancreas transplantation. The benefits of pancreas transplantation are 1. freedom from insulin therapy and the metabolic derangements of type 1 diabetes mellitus and 2. potential slowing or reversal of the progression of end-organ damage from this condition. Disadvantages include higher risk of surgical complications and the need for higher levels of immunosuppression. Although there is growing evidence to support kidney-pancreas transplantation in those with type 2 diabetes, the vast majority are performed in individuals with type 1 diabetes. Options include simultaneous kidney and pancreas (SPK) transplantation or pancreas-after-kidney transplantation (the latter allows living donor kidney transplantation). The half-life of SPK pancreas grafts is 14 years.[15]

Pancreas transplant alone can be performed in those without kidney failure, though it is unclear if pancreas transplant alone or pancreas after kidney transplantation improves patient survival (worse pancreas allograft outcomes compared with SPK). The major causes of graft failure are technical (40%) and acute rejection (15%) in the early posttransplant period and chronic rejection (25%) in the late posttransplant period.[16] Rates of pancreas transplantation have decreased somewhat over the past decade from a peak of ~1200 procedures in 2010 to ~1000 in 2021 as the management of type 1 diabetes has improved with closed loop systems, etc.

Complications of pancreatic transplantation include thrombosis, infection, rejection, and problems related to drainage of the exocrine secretions.[17] Drainage of exocrine secretions into the bladder (compared with enteric drainage) affords the advantages of sterility and of serial measurement of urinary amylase concentrations that can aid in early detection of pancreatic allograft dysfunction. Important disadvantages include severe cystitis, hypovolemia, and acidosis (the last two due to large losses of bicarbonate-rich fluid).

ASSESSMENT OF COMPATIBILITY AND IMMUNOLOGIC RISK

SOLID-PHASE TECHNOLOGIES

The presence of preformed recipient antibody against either the donor ABO blood group or donor HLA may result in hyperacute rejection. Solid-phase technologies such as the Luminex single-antigen bead (SAB) anti-HLA antibody assay permit detection and quantification of a wide array of anti-HLA donor-specific antibodies (DSAs) (discussed further in Chapter 68). Multiple color-coded beads coated with a broad range of HLA antigens are incubated with recipient serum. Binding of recipient antibody to a bead coated with a specific HLA antigen results in a distinct fluorometric signal, which is then detected by flow cytometry. Molecular (polymerase chain reaction) HLA typing techniques have now superseded serologic identification of HLA type for both donors and recipients. Accurate molecular donor HLA typing and SAB-based recipient anti-HLA antibody screening can be combined to perform a virtual crossmatch. For high-immunologic-risk donors, virtual crossmatching has assisted in the identification of potentially compatible donors and has improved transplant rates for sensitized donors.[18] Computerized virtual crossmatch algorithms have also been central to the success of kidney paired donor exchange.[19,20] In the United States, wait-listed kidney transplant candidates usually have SAB-based HLA-antibody screening performed every 3 months. HLA antigens against which a candidate has antibodies are listed as unacceptable HLA antigens.

In 2009, the calculated panel reactive antibody (cPRA) replaced the measured PRA as the main measure of immune sensitization. Sensitizing events include transfusions, pregnancy, and prior transplants. The cPRA is calculated using an algorithm that correlates donor anti-HLA antibody (as measured by SAB assay) with the population frequency of HLA antigens to generate a percent score. A score of 0% suggests no anti-HLA antibodies, while a sensitized candidate with a cPRA of 85% will be immunologically incompatible with 85% of the donor population.

DSA may be present pretransplant or develop de novo post transplant. De novo DSA development frequently follows T cell–mediated rejection (TCMR) and is also associated with patient nonadherence.[21] Both preformed and de novo DSA are associated with an increased risk of antibody-mediated rejection (ABMR) and poor graft outcomes.[22,23] DSAs may also be measured using a variation on the SAB technique, the Luminex C1q assay, which detects DSA that bind and activate complement. The development of de novo C1q-binding DSA is strongly associated with poor graft outcome.[24] DSA IgG subtyping also yields similarly helpful prognostic information, with DSAs of the IgG3 subclass being associated with the worst clinical prognosis.[25]

HUMAN LEUKOCYTE ANTIGEN EPITOPE MATCHING

Though not widespread in clinical use, HLA epitope matching is performed by separating HLA antigens into a series of epitopes—regions that are capable of binding antibody (discussed further in Chapter 68). HLA epitopes may be shared across different HLA molecules. The regions of the epitope that are targets for antibody binding are short, polymorphic, often nonlinear amino acid sequences on the HLA molecule that are clustered together in the three-dimensional structure of the HLA molecule and known as "eplets."[26] Donor-recipient compatibility is then expressed as the number of eplet mismatches. Minimizing donor-recipient epitope mismatch reduces the number of viable foreign targets to which the recipient may make antibodies and represents a more relevant way to quantify tissue mismatch.

COMPLEMENT-DEPENDENT CYTOTOXICITY AND FLOW CROSSMATCH ASSAYS

The complement-dependent cytotoxicity (CDC) assay, which incubates donor lymphocytes with recipient serum, was developed in the 1960s.[27] Preformed recipient antibody to donor lymphocytes (subsequently identified as anti-HLA antibody) resulted in lymphocyte death in vitro and predicted rapid (hyperacute) graft failure in vivo. The test is now performed as two separate donor T cell (express HLA class I antigen) and B cell assays (express HLA class II antigen). A positive CDC T cell crossmatch is an absolute contraindication to transplantation. An isolated positive CDC B cell crossmatch may indicate the presence of preformed antibody to class II HLA antigen, low-level antibodies against class I antigen, or non-HLA antibodies.

The flow crossmatch (FCXM) is a more sensitive assay for detection of donor anti-HLA antibody. Donor T cells or B cells are mixed with recipient serum and a fluorescent anti-IgG antibody. Any *recipient* antibody that binds to donor HLA will be tagged by the fluorescent IgG and subsequently detected using flow cytometry. FCXM can detect low-titer and non–complement-binding anti-HLA antibodies and may be positive when a CDC crossmatch is negative. A negative CDC crossmatch but positive T cell FCXM is associated with poor short-term transplant outcomes and is a relative contraindication to transplantation.[28]

THE TRANSPLANT SURGERY PROCEDURE AND SURGICAL COMPLICATIONS

Before the kidney allograft is placed in the extraperitoneal iliac fossa (Fig. 69.1), a curvilinear incision is made in either lower quadrant, the retroperitoneal space is widened, and iliac vessels are exposed. The external iliac artery and vein are mobilized, and surrounding lymphatic vessels are ligated and divided. Depending on the length, size, and quality of the vessels, end-to-side anastomoses are performed between 1. the donor renal vein and recipient external iliac vein, and 2. the donor renal artery and donor external iliac artery (or common or internal iliac artery). A third and final connection is the implantation of the ureter to the bladder (neoureterocystostomy). Prophylactic ureteral stents have been shown to lower the risk of urologic complications including ureteric stenosis and urinoma.[29]

Advances in donor and recipient surgical technique, anesthesia, organ preservation, and perioperative care have all likely contributed to improvements in 1-year patient and graft survival rates.[30] Early recognition and management of surgical complications remain important. These may include hemorrhage, renal artery or vein thrombosis, arterial anastomosis pseudoaneurysm, lymphocele, and urinomas (Table 69.1).

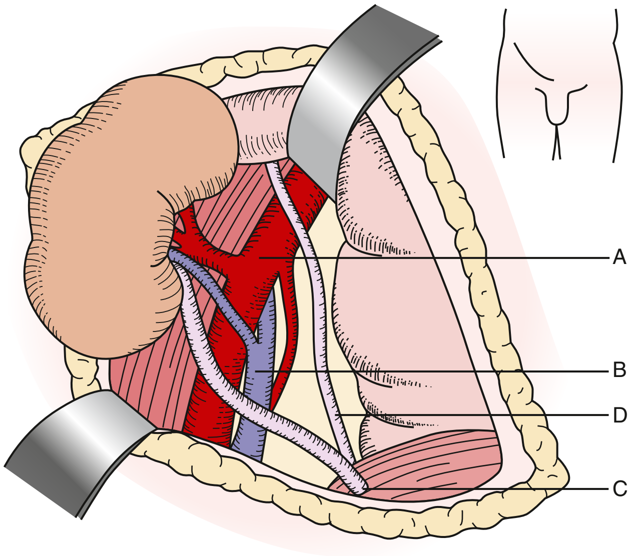

Fig. 69.1 Anatomy of a typical first kidney transplant. (A) External iliac artery, (B) external iliac vein, (C) implanted donor ureter, and (D) native ureter.

DECEASED DONOR KIDNEY RETRIEVAL AND PRESERVATION

The goal of organ preservation is to prevent damage from ischemia by 1. retrieving and cooling the organs expediently, and 2. replacing the circulating blood with a preservation solution, which is designed to minimize intracellular edema, preserve the integrity of cells and tissue, and buffer free radicals. The two cold storage preservation solutions most widely used in the United States are the University of Wisconsin (Viaspan, UW) and Histidine-Tryptophan Ketoglutarate (Custodiol, HTK) solutions (**eTable 69.1**). The UW solution had been the preservation solution of choice for multiorgan deceased donor recovery, though HTK use is increasing in a number of Organ Procurement Organizations. Two small randomized controlled trials (RCTs) comparing UW with HTK preservation solutions found no difference in DGF or graft survival between groups.[31,32] A large retrospective (nonrandomized) study comparing HTK to UW preservation solution found that HTK use was associated with an increased risk of late death-censored graft loss.[33] For living donor kidneys with short

Table 69.1 Surgical Complications of the Transplant Procedure[149,413-415]

Complication	Description	Management
Hemorrhage	Usually involves the anastomotic site. Perirenal hematomas can result from venous or small artery bleeding.	Unless small and stable, surgical exploration is required. Large hematomas should be evacuated in order to reduce the risk for subsequent infection.
Renal artery thrombosis	Rare (<1%) and usually related to an anastomotic problem or kink in the renal artery. Risk factors include recipient arteriosclerosis, multiple donor arteries, vasospasm, and hypotension.	Delays inherently associated with confirming the diagnosis and preparing the patient for surgical exploration usually exceed the time required to reestablish arterial flow to the kidney, resulting in prolonged warm ischemia, hypoxia, and often permanent loss of function.
Renal vein thrombosis	Causes include problems with the surgical anastomosis, extrinsic compression (by a lymphocele/hematoma), or deep venous thrombosis that extends in the iliac vein at the level of the venous anastomosis; thrombophilia may contribute. Allograft ultrasound shows decreased or absent blood flow in the renal vein and either absent/reversed diastolic arterial flow.	Surgical exploration to attempt thrombectomy followed by anticoagulation can be performed but is rarely successful.
Arterial anastomosis pseudoaneurysm	Rare, infectious complication that leads almost invariably to graft loss, associated with significant mortality and morbidity. Though rare, distal thrombosis of the femoral artery or arterial dissection can lead to loss of a lower limb.	Transplant nephrectomy, vascular reconstruction and/or excision with extra anatomic bypass is usually required. If less severe, treatment with covered stenting can be attempted.
Lymphocele	Lymphatic fluid collection originating from either the severed iliac lymphatics or lymphatic drainage of the renal allograft itself. Analysis of aspirated fluid will typically show a high lymphocyte count and a creatinine concentration similar to serum.	Lymphoceles frequently reaccumulate following drainage, although injection of sclerosing agent following aspiration may reduce the risk of recurrence. The preferred and more definitive treatment is internal drainage of the lymphocele into the peritoneal cavity. In many centers, a laparoscopic transabdominal approach has replaced the traditional open approach that utilizes the kidney transplant incision.
Urinoma/urine leak	Causes include infarction of the ureter due to perioperative disruption of its blood supply (typically a lower pole artery infarction) and breakdown of the ureterovesical anastomosis. Analysis of aspirated fluid will typically show a fluid creatinine concentration much higher than serum.	A bladder catheter should be immediately inserted to decompress the urinary tract. Many cases require surgical exploration and repair. The type of repair depends on the level of the leak and viability of involved tissues.

eTable 69.1 Common Cold Storage Preservation Solutions

Solution	Minimization of Intracellular Edema
UW	Viscosity 3x water; composition similar to intracellura fluid (potassium meq/L); lactobionate and raffionose prevent cellular edema and hydroxyethyl first buffers free radicals.
HTK	Low viscosity, low sodium and potassium (Both 15 mmol/L); Histidine serves buffer; Tryptophan and mannitol are free radical scavengers; low cost

UW, University of Wisconsin; *HTK,* Histidine-Tryptophan-Ketoglutarate.

ischemia times, the use of simple and inexpensive solutions such as heparinized lactated Ringer with procaine has been proposed.

Hypothermic machine perfusion is an alternative to static cold preservation for deceased donor kidneys. Following standard retrieval and flushing of the kidney allograft, the renal artery is connected to a perfusion pump that circulates a preservation solution, maintained at 1°C to 10°C. Perfusion parameters can be used to determine the quality of the graft. Hypothermic machine perfusion is associated with a reduction in DGF compared with static cold storage, and some studies have shown improvements in 1-year graft survival.[34] There is increasing interest in ex vivo normothermic perfusion as a preservation method; one potential advantage of this technique is that it allows ex vivo assessment of graft function.[35]

CURRENTLY USED IMMUNOSUPPRESSIVE AGENTS IN KIDNEY TRANSPLANTATION

OVERVIEW AND THREE-SIGNAL T CELL ACTIVATION

Immunosuppressive therapy, generally described as either induction or maintenance, aims to deplete lymphocytes or act on downstream T cell activation signals. Induction is defined as the rapid achievement of profound immunosuppression at the time of transplant. Maintenance immunosuppression is achieved by combining agents that take advantage of additive or synergistic immunosuppressive effects of different drug categories while minimizing their adverse effects. The immunosuppressive drugs commonly used in clinical transplantation, along with adverse effects, are summarized in Table 69.2 and **eTable 69.2**, and their mechanisms of action are illustrated in Fig. 69.2. A combination of a calcineurin inhibitor (CNI) and antiproliferative agent is the most common regimen, with the addition of corticosteroids depending on the transplant center's protocol, risk of rejection, and patient characteristics.

T cells play a central role in the recognition of the allograft as foreign and initiation of the rejection process. The T-cell immune response requires three distinct signaling events (see Fig. 69.2): 1. Antigen-presenting cells (APCs: dendritic cells, macrophages, activated endothelium), most often of donor origin (direct allograft recognition), migrate to the recipient's secondary lymphoid organs where foreign antigen/major histocompatibility (MHC) complex is presented to the recipient's T cell receptor (TCR) (Signal 1). In indirect or semidirect allorecognition, *recipient* APCs present antigens to recipient T cells. 2. During the next costimulatory step (Signal 2), APC B7 (CD80 and CD86) engage with CD28 on T cells. 3. Signals 1 and 2 lead to the activation of 3 pathways: Calcium–calcineurin, the RAS–mitogen-activated protein (MAP) kinase, and the nuclear factor-kB. Activation of these pathways results in cytokine production (e.g., interleukin-2 [IL-2], interleukin-15) and expression of surface molecules (e.g., IL-2 receptor, CD25). The cytokines then stimulate T cell proliferation (Signal 3).[36]

The B cell immune response is increasingly being targeted by newer therapies in kidney transplantation. B cell antigen receptor engagement with antigens activates B cells and ultimately the production of donor-specific anti–human

Table 69.2 Drugs Used in Maintenance Immunosuppression

Drug	Mechanism of Action	Adverse Effects
Corticosteroids	Block synthesis of several cytokines including IL-2; multiple antiinflammatory effects	Glucose intolerance, hypertension, hyperlipidemia, osteoporosis, osteonecrosis, myopathy, cosmetic defects, growth suppression in children
Cyclosporine	Inhibits calcineurin (binds cyclophilin)-dependent synthesis of IL-2 and other molecules critical for T cell activation; T cell activation thereby inhibited	Nephrotoxicity (acute and chronic), hypertension (more than tacrolimus), posttransplant diabetes, neurotoxicity, hyperkalemia, hypomagnesemia, metabolic acidosis, hypercalciuria, and dyslipidemia (more than tacrolimus). Hypertrichosis, gingival hyperplasia (particularly when used in combination with dihydropyridine calcium channel blockers)
Tacrolimus	Similar to cyclosporine, though binds FKBP	Similar to cyclosporine, though more hyperkalemia, neurotoxicity, posttransplant diabetes, and alopecia
Azathioprine	Inhibits purine biosynthesis; lymphocyte replication therefore inhibited	Bone marrow suppression, rarely pancreatitis, hepatitis
Mycophenolate mofetil	Inhibits de novo pathway of purine biosynthesis (relatively lymphocyte selective); lymphocyte replication therefore inhibited	Bone marrow suppression, gastrointestinal upset, invasive CMV disease more common than with azathioprine
Sirolimus	Sirolimus-FKBP complex inhibits TOR blocking lymphocyte proliferative response	Bone marrow suppression, proteinuria, mouth ulcers, hyperlipidemia, interstitial pneumonitis, edema, enhanced nephrotoxicity of cyclosporine/tacrolimus
Belatacept	Blocks T cell costimulation	PTLD in EBV seronegative, PML (rare), reactivation of TB

CMV, Cytomegalovirus; *EBV,* Epstein-Barr virus; *FKBP,* FK-binding protein; *IL-2,* interleukin-2; *PTLD,* posttransplant lymphoproliferative disorder; *TB,* tuberculosis; *TOR,* target of rapamycin.

eTable 69.2 Toxicity Profiles of Immunosuppressive Medications

		Calcineurin Inhibitors		Antiproliferative Agents		
Adverse Effect	**Steroids**	**CsA**	**Tac**	**MMF**	**AZA**	**mTORi**
Posttransplant diabetes	↑	↑	↑↑			↑
Dyslipidemia	↑	↑				↑↑
Hypertension	↑↑	↑↑	↑			
Osteopenia	↑↑	↑	(↑)			
Anemia and leukopenia				↑	↑	↑
Delayed wound healing						↑
Diarrhea, nausea/vomiting			↑	↑↑		↑
Proteinuria						↑↑
Decreased eGFR		↑↑	↑			
Hair changes		↑	↓			
Gingival hyperplasia		↑				
Peripheral edema	↑					↑

AZA, Azathioprine; *CsA,* cyclosporine A; *MMF,* mycophenolate mofetil; *mTORi,* mammalian target of rapamycin inhibitor(s); *Tac,* tacrolimus.
↑ Indicates a mild-moderate adverse effect of the complication.
↑↑ Indicates a moderate-severe adverse effect of the complication.
(↑) Indicates a possible, but less certain adverse effect on the complication.
Adapted from KDIGO clinical practice guideline for the care of kidney transplant recipients. *Am J Transplant.* 2009;9(Suppl 3):S1–155.

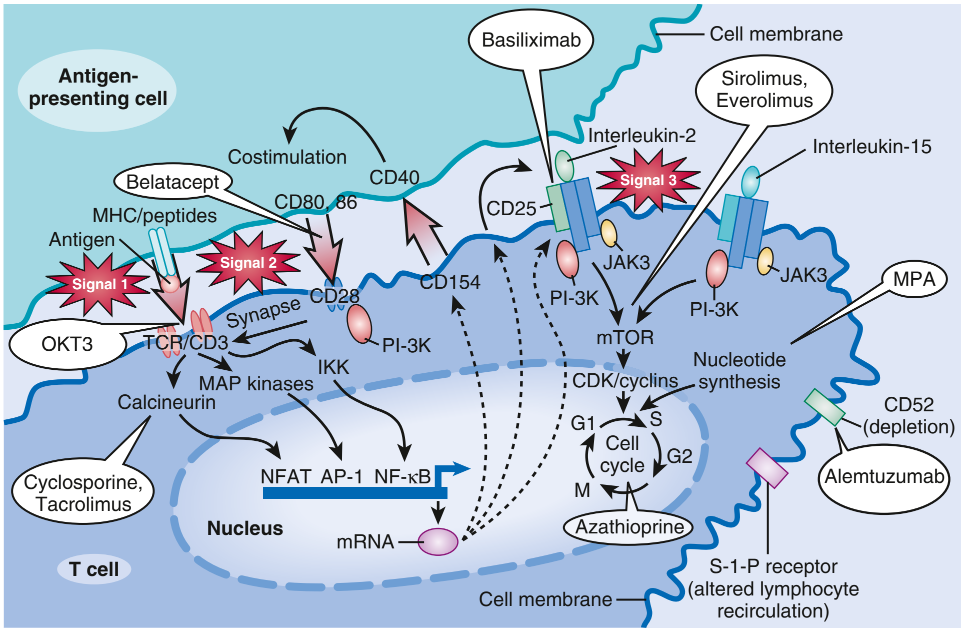

Fig. 69.2 Stages of T-cell activation: multiple targets for immunosuppressive agents. Signal 1: The Ca^{++}-dependent signal induced by T cell receptor (TCR) stimulation results in calcineurin activation, a process inhibited by the calcineurin inhibitors (CNIs). Calcineurin dephosphorylates nuclear factor of activated T cells (NFAT), enabling it to enter the nucleus and bind to the interleukin-2 (IL-2) promoter. Corticosteroids bind to cytoplasmic receptors, enter the nucleus, and inhibit cytokine gene transcription in both the T cell and antigen-presenting cell (APC). Corticosteroids also inhibit activation of the transcription factor, nuclear factor-κB (not shown). Signal 2: Costimulatory signals, such as that between CD28 on the T cell and B7 on the APC, are necessary to optimize T cell transcription of the IL-2 gene, prevent T-cell anergy, and inhibit T cell apoptosis. Signal 3: IL-2 receptor stimulation induces the cell to enter the cell cycle and proliferate. IL-2 and related cytokines have both autocrine and paracrine effects. Signal 3 may be blocked by IL-2 receptor antibodies or by sirolimus. Further downstream, azathioprine and mycophenolate mofetil (MMF) inhibit progression into the cell cycle by inhibiting purine and therefore DNA synthesis. (From Halloran PF. Immunosuppressive drugs for kidney transplantation. *N Engl J Med.* 2004;351:2715–2729.)

leukocyte antigen (HLA) antibodies and differentiation into antibody-producing plasma cells. Long-lasting immune memory can develop through the generation of memory B cells.[37] The immunobiology of transplantation is discussed in detail in Chapter 68.

INDUCTION THERAPIES

Depending on a kidney transplant recipient's risk of rejection, balanced with that of infection and malignancy, several induction therapy options are available and are summarized as follows.

MONOCLONAL INTERLEUKIN-2 RECEPTOR ANTAGONIST

Basiliximab is a chimeric monoclonal antibody against the α chain of the T cell IL-2 receptor (CD25), which provides prophylaxis for 30 days post transplant. Compared with placebo, it reduces rejection rates approximately 30% to 40%.[38-40] As it is not a lymphocyte-depleting agent and acts on downstream T cell activation pathways, infection risk is lower compared with the lymphocyte-depleting agents rabbit ATG and alemtuzumab.[41-43] Compared with polyclonal lymphocyte-depleting therapies, injection-related side effects and risk of infection and cancer are minimal.[39,40] Basiliximab is administered as two 20 mg doses, on the day of transplantation and postoperative day 3 or 4.

POLYCLONAL LYMPHOCYTE-DEPLETING AGENTS

Polyclonal antibodies against lymphocytes are prepared by inoculation of animals with human lymphoid cells. The first clinical use in the 1970s of a polyclonal lymphocyte-depleting agent was that of ATG derived from horse inoculation (hATG, Atgam). Rabbit ATG (rATG, Thymoglobulin) is derived from rabbit inoculation with human lymphoid cells, which results in production of antibodies that target more than 20 different T cell epitopes.[44,45] T cell depletion is thought to involve complement-dependent lysis, apoptosis, and phagocytosis in the peripheral lymphoid tissue. Antibodies against adhesion molecules that are present in ATG preparations may also play a role by modulating leukocyte function.[46] Horse ATG is now rarely used as initial induction therapy, though it may be considered in individuals with a hypersensitivity reaction to rATG or significant pretransplantation rabbit exposure. A 2008 randomized trial that compared hATG with rATG induction therapies demonstrated rATG superiority in terms of reduction of the composite of death, graft loss, or rejection.[47]

Several RCTs have compared ATG induction with other strategies such as IL-RA, alemtuzumab, and OKT3 (a murine monoclonal anti-CD3 antibody no longer used to due to adverse effects). Induction with rATG resulted in equivalent 1-year graft survival but a lower rate of acute rejection, as well as both infection and non–infection-related complications compared with OKT3 in patients treated with cyclosporine, azathioprine, and steroid maintenance immunosuppression.[48] In sensitized patients, defined as current or peak panel reactive antibodies (PRA) >30% or >50%, respectively, rATG induction was associated with significantly lower rates of biopsy proven acute rejection and steroid-resistant rejection in the first year post transplant, without any difference in 1-year survival compared with daclizumab (IL-2 receptor agonist [RA] no longer used) induction.[41] Multiple clinical studies have compared rATG with other agents.

A 1981 study that compared hATG induction with no induction in patients treated with azathioprine and steroid-based maintenance immunosuppression regimen found a significantly improved 2-year graft survival in the hATG-treated group.[49] In another study of kidney transplant recipients with risk factors for delayed graft function (DGF) or acute rejection, rATG resulted in a lower acute rejection rate than basiliximab induction, but no difference in DGF or 1-year graft survival. Five-year follow-up of this study showed sustained superiority of rATG in terms of acute rejection.[42,50] Notably, infectious complications were more common in the recipients of rATG (vs. IL-2 RA) in both trials. In immunologically high-risk kidney transplant recipients randomly assigned to either rATG or alemtuzumab (see later) induction, no differences were seen in acute rejection rates between the groups followed out to 3 years post transplant,[43] although infectious complications were more common in rATG-treated subjects. A 2017 retrospective analysis of U.S. kidney transplant recipients that compared outcomes in matched patients receiving induction with rATG versus alemtuzumab or basiliximab found that rATG use was associated with improved outcomes: death or allograft failure versus alemtuzumab, death versus basiliximab.[51]

ATG also has an important role in the treatment of severe or steroid-resistant acute cellular rejection. A randomized study showed rATG superior to hATG in the treatment of acute rejection in terms of rejection reversal and 90-day recurrence rates.[52]

ATG increases the risk of malignancy when compared with IL-2 RA induction therapy, and greater exposure may lead to greater incidence of malignancy.[53] Infusion-related side effects, such as fever, chills, hypotension and, less frequently, cardiovascular events, are usually mild, particularly with adequate steroid and antihistamine premedication and a slow infusion rate. These reactions are more likely to occur during the first few infusions and become rare with subsequent infusions. Cytokine release syndrome, a systemic inflammatory response with high interleukin-6 (IL-6) levels, has been reported after ATG administration and can present with mild, flulike symptoms or severe life-threatening manifestations.[54] Serum sickness, characterized by fever, rash, and arthralgia occurring 10 to 15 days after treatment, has also been reported, possibly more frequently in patients not receiving steroid prophylaxis.[55]

MONOCLONAL ANTI-CD52 ANTIBODY (ALEMTUZUMAB, DEPLETING)

Alemtuzumab, a humanized monoclonal antibody against CD52, which was originally developed to treat refractory B cell chronic lymphocytic leukemia, depletes both T and B cells. In an RCT that compared alemtuzumab with induction with basiliximab (immunologically low risk) and rATG (high risk), alemtuzumab was superior to basiliximab and equivalent to rATG for acute rejection up to 3 years.[43] Development of autoimmune disease has been reported in solid organ transplant recipients treated with alemtuzumab induction.[56,57] One limitation to its use is the lack of wide commercial availability. It is typically dosed as a one-time 30-mg injection.

MONOCLONAL ANTI-CD20 ANTIBODY (RITUXIMAB, OBINUTUZUMAB)

Rituximab is a chimeric anti-CD20 cytolytic monoclonal antibody used as induction therapy for immunologically high-risk patients based on evidence that it has a favorable safety profile and is associated with a reduction in rejection and antibody-mediated graft dysfunction.[60] In patients at high risk for ABMR, rituximab may be used in conjunction with intravenous immunoglobulin (IVIG, see later).[61,62] Hepatitis B prophylaxis should be initiated in patients with prior hepatitis B infection. Rituximab may also be used for desensitization (see later)[63,64] or treatment of recurrent or de novo posttransplant glomerular diseases (e.g., membranous nephropathy, focal segmental glomerulosclerosis [FSGS]).[65,66] Premedication with steroid, acetaminophen, and an antihistamine is given to avoid an infusion reaction. Obinutuzumab, a newer anti-CD20 monoclonal antibody directed at a different CD20 epitope, has been shown to effectively deplete B cells in kidney transplant recipients.[67]

INTRAVENOUS IMMUNOGLOBULIN

Various pooled human IVIG formulations are available to use before transplantation as part of both HLA- or ABO-incompatible desensitization protocols, as part of induction therapy in patients with preformed DSAs or for treatment of ABMR.[58,68-70] While its mechanism of action is not fully understood, it is thought to involve: 1. neutralization of circulating antibody, 2. complement inhibition, 3. modulation of B cell and APC function, and 4. cytokine inhibition.[71-73] Adverse effects include infusion reaction, headache, aseptic meningitis, hemolysis (especially in patients who are blood group A) and, rarely, thrombosis. IVIG can also be used to treat posttransplant viral infections such as BK virus (discussed later) and parvovirus, a viral infection associated with severe anemia and proteinuria.[70]

DESENSITIZATION

Desensitization protocols attempt to lower anti-HLA antibody titers to create an immunologic window for successful transplantation in highly sensitized patients and are associated with good medium-term graft survival rates, as well as a survival advantage compared with remaining on dialysis.[63,74,75] Unfortunately, ABMR rates are high[63,76] and few data are available about long-term graft survival in these patients.

Desensitization protocols fall into two broad categories: 1. high-dose IVIG/anti-CD20-based[64] and 2. low-dose IVIG/plasmapheresis-based[74] regimens. A number of studies have suggested that IVIG without rituximab or plasmapheresis is associated with a posttransplant rebound in DSA titers and increased incidence of ABMR.[77] More recent reports have described successful desensitization using newer agents such as the anti-IL-6 receptor antibody (tocilizumab) and the IgG-degrading enzyme (IdeS).[78,79]

For living donor candidates, a donor exchange program can avoid the need for desensitization.[80] The new U.S. Kidney Allocation System (KAS), which offers significant allocation score bonuses and high priority access to kidney transplants across the country for the most sensitized patients on the list (calculated PRA >98%), was implemented in December 2014. KAS has resulted in a significant increase in transplantation rates for highly sensitized patients.[81,82]

ABO-INCOMPATIBLE TRANSPLANTATION

An important immunologic barrier in kidney transplantation is ABO blood group status, as blood group antigens are expressed on endothelial cells. With certain exceptions, ABO-incompatible transplantation without desensitization will result in ABMR.[83] In patients with low baseline anti-A/B titers, ABO-incompatible transplant has been shown to be safe with minimal pretreatment.[84] Blood group A donors are composed of individuals with different A subtypes. The most common are type A1 and A2; A2 antigens are less highly expressed than A1 antigens. The transplantation of blood group A2 donors into B recipients with low anti-A titers is considered safe without desensitization.[84-86] KAS has permitted that A2 or A2B deceased donor kidneys be allocated to selected blood group B candidates to reduce the long wait times faced by this group.

Much of the pioneering work on desensitization for ABO-incompatible transplant has been done in Japan, where deceased donor transplants are not performed.[87] ABO-incompatible desensitization protocols mimic those described earlier, with a goal to lower anti-A/B titers to under 1:8 or 1:16. Some protocols also include posttransplant plasmapheresis or IVIG.[88-90] The long-term outcome of desensitized recipients of blood group incompatible organs is favorable, although perhaps slightly inferior to that of ABO-compatible kidney transplant recipients.[91,92]

MAINTENANCE IMMUNOSUPPRESSIVE AGENTS

CALCINEURIN INHIBITORS

Calcineurin is a calcium-dependent serine/threonine phosphatase that is involved in a diverse range of cellular functions including T cell signal transduction.[93] Signal 2 (described earlier) activates the calcium-calcineurin pathway, triggering cytosolic influx of calcium and the downstream activation of calcineurin. Activated calcineurin dephosphorylates the transcription factor nuclear factor of activated T cells (NFAT), which then allows its translocation to the nucleus. In the nucleus, NFAT activates a host of target genes including the cytokine IL-2.[94] IL-2 then binds to its receptor and initiates T cell expansion.[36] The excellent posttransplant outcomes seen with the advent of the CNIs in the 1980s made the widespread expansion of solid organ transplantation possible. CNIs, first in the form of cyclosporine (CsA, binds the immunophilin cyclophilin) and then later tacrolimus (FK, binds the immunophilin FK binding protein 12), have remained a backbone of maintenance immunosuppression regimens. Pharmacokinetic monitoring is most often done by measuring a 12-hour trough level (C_0) of CsA and FK, a reasonable proxy for area under the curve for clinical use.[95]

As the cytochrome P450 enzyme system (CYP3A) is responsible for CNI metabolism, inducers (e.g., rifampin, phenobarbital, phenytoin) or inhibitors (e.g., ketoconazole, erythromycin, ritonavir, diltiazem, cannabidiol) of this enzyme can decrease or increase CNI levels, respectively (Table 69.3). Polymorphisms in *CYP3A5* and *ABCB1* (encodes efflux transporter P-glycoprotein present in enterocytes) can also impact CNI levels.

Unfortunately, CNIs are not without side effects including CNI-associated nephrotoxicity, which may contribute to a reduction in long-term graft survival in a proportion of kidney transplant recipients (**eTable 69.2**).[96-99] CNI nephrotoxicity may present as diverse kidney pathology including thrombotic microangiopathy, striped interstitial fibrosis, and arteriolar hyalinosis. Other side effects include hypertension (more with CsA), post-transplant diabetes (more with FK), neurotoxicity (more with FK), hyperkalemia (more with FK), hypomagnesemia, metabolic acidosis, hypercalciuria, and dyslipidemia (more with CsA). FK and CsA can have opposing effects on hair growth, with FK associated with alopecia and CsA with hypertrichosis. CsA may cause gingival hyperplasia, particularly when used in combination with dihydropyridine calcium channel blockers.[95] The overall contribution of CNI-specific chronic injury, however, remains controversial in the landscape of current therapy. Concern regarding CNI nephrotoxicity has led to a greater focus on strategies to minimize CNI exposure. Data from two RCTs from the late 2000s supported the efficacy and safety of lower-dose cyclosporine and tacrolimus-based regimens in immunologically low-risk patients.[100,101] However, antibody-mediated allograft injury has been increasingly recognized as an important cause of late graft loss.[102] CNI underexposure may allow for increased allo-recognition, chronic ABMR, and higher risk of new DSA formation.[103-106] To date, belatacept (see later) probably represents the best available option for calcineurin avoidance.[107]

Cyclosporine (CsA)

Both European and American clinical trials have demonstrated the superiority of cyclosporine either alone or with steroid over the standard immunosuppression regimen of azathioprine and steroid (from the 1960s).[108,109] The oil-based formulation (Sandimmune) is associated with erratic gastrointestinal absorption and highly variable bioavailability, while microemulsion formulations (Neoral, Gengraf) improve drug absorption and the pharmacokinetic profile.[110,111]

Tacrolimus (FK)

Trials studying the efficacy of FK have demonstrated reduced rates of rejection compared with cyclosporine,[112] particularly the oil-based formulation of cyclosporine. Mycophenolic acid exposure is reduced by approximately 40% when combined with cyclosporine versus tacrolimus, an effect that likely contributes to the relatively greater immunosuppressive potency of tacrolimus compared with cyclosporine-based CNI-MMF regimens.[113] Extended-release formulations of tacrolimus have been approved in the United States and may be beneficial in terms of adherence and neurotoxicity mitigation.[114-116]

Table 69.3 Agents That May Interact With Transplant Medications

Class of Drug	Increase Level	Decrease Level
Drugs That Interact With Calcineurin Inhibitors and Sirolimus		
Ca channel blocker	Diltiazem, verapamil	
Antibiotics	Erythromycin, azithromycin, clarithromycin	Nafcillin
Antifungals	Fluconazole, ketoconazole, itraconazole, voriconazole	
Antituberculin		INH, rifampin, rifabutin
Antiviral	Ritonavir, nelfinavir, saquinavir	Efvirez, nevirapine
Antiseizure		Phenytoin, phenobarbital, carbamazepine, primidone
Antidepressant	Fluoxitine, nefazodone, fluvoxamine	
Foods and Herbal Preparations That Interact With Calcineurin Inhibitors		
Food	Grapefruit juice/pomegranate juice	
Herbs		St. Johns wort

ANTIPROLIFERATIVE AGENTS

Azathioprine

Antiproliferative agents, as the name implies, interrupt T cell proliferation. Azathioprine (Imuran) is a prodrug whose metabolites: 1. incorporate thioguanine purine analogs into DNA and RNA resulting in cell death, 2. inhibit de novo purine synthesis by methylthioinosine monophosphate, and 3. inhibit the Rho family GTPase, Rac1, which leads to T cell apoptosis.[117] Azathioprine has now been largely superseded by mycophenolate mofetil (MMF),[118] although longer vintage kidney transplant recipients may remain on it. Azathioprine is commonly substituted for MMF in patients with kidney transplants who are planning to conceive or who have MMF-related gastrointestinal intolerance. An important potential adverse effect of azathioprine is myelosuppression. Individuals with reduced thiopurine methyltransferase enzyme activity (TPMT) tend to accumulate active drug metabolites and are predisposed to hematologic toxicity.[119] Notably, coadministration of allopurinol with azathioprine also shifts azathioprine metabolism toward the production of metabolically active substrates and may lead to potentially serious toxicity.

Mycophenolic acid

MMF and mycophenolate sodium (Myfortic) are two formulations of mycophenolic acid (MPA), an inhibitor of inosine monophosphate dehydrogenase (IMPDH). IMPDH is required for the de novo purine synthesis pathway of guanosine from inosine. The effects of MPA are relatively lymphocyte specific, as these cells lack a purine salvage pathway. Thus MPA prevents T and B lymphocyte replication and suppresses both the cellular and humoral immune responses. Clinical trials have demonstrated that MMF reduces acute rejection rates by 50% compared with azathioprine or placebo.[120] A meta-analysis of 19 trials found that MMF was associated with a reduction of acute rejection and improved graft survival compared with azathioprine.[118]

Drug-drug interactions may impact MPA absorption. MMF (unlike mycophenolate sodium) requires an acidic gastric environment for cleavage and thus proton pump inhibitors, which raise the gastric pH, can impair drug absorption.[121] Cyclosporine blocks the enterohepatic recirculation of MMF, reducing the exposure of the drug by approximately 40% compared with using tacrolimus.[113] MMF can cause gastrointestinal side effects, neutropenia, and, less frequently, anemia, which may require dose reduction or stoppage. Alopecia may also occur with MMF treatment.

Despite the high interindividual and intraindividual variability of its pharmacokinetics, routine therapeutic drug monitoring of MMF has not been established. MMF is approved at a dose of 1 g twice daily when used in conjunction with CNI and steroids. The excellent results obtained with a standard dosing regimen and the poor correlation of a single time point to AUC (particularly when used in combination with CsA) support a fixed-dose regimen.[36,122]

Mycophenolate sodium is an enteric-coated slow-release MPA formulation, absorbed in the small intestine, that was developed to decrease the gastrointestinal side effects of MMF. Though the data are mixed, switching from MMF to the enteric-coated formulation may improve gastrointestinal symptom burden.[123,124] The dose conversion for MMF to mycophenolate sodium is 250 mg to 180 mg. MPA is associated with an increased risk of major fetal malformations, and patients who are planning pregnancy should be switched to azathioprine at least 6 weeks before attempting to conceive.[125]

MAMMALIAN TARGET-OF-RAPAMYCIN INHIBITORS

Sirolimus (Rapamune), a macrocyclic antibiotic with immunosuppressive properties, and Everolimus (Zortress), derived from sirolimus and with a shorter half-life, are examples of mammalian target-of-rapamycin (mTOR) inhibitors. Even though it shares its cytoplasmic binding protein with FK, the resulting complex does not interfere with calcineurin and instead binds mTOR. mTOR inhibition prevents the propagation of IL-2-mediated cell proliferation signaling through the PI3K/AKT/mTOR pathway.[126] Sirolimus inhibits cellular proliferation of T and B lymphocytes and reduces antibody production.[36] Its immunosuppressive effects have been demonstrated in preclinical studies to be synergistic with CsA and

FK.[127,128] Side effects of sirolimus include hyperlipidemia, thrombocytopenia, anemia, poor wound healing, diarrhea, pneumonitis, mouth ulcers, proteinuria, and peripheral edema.

mTOR inhibitors are not routinely used as part of maintenance immunosuppression regimen, as clinical studies have shown inferior outcomes (increased acute rejection, lower GFR) when compared with CNI. Sirolimus-related adverse events and drug discontinuation have also been a challenge in clinical studies.[129] In addition, a trend toward worse GFR in CNI and sirolimus treatment combinations (especially cyclosporine) is consistent with animal studies that show that sirolimus cotreatment potentiates CNI nephrotoxicity.[130-133]

Late conversion from CNI to sirolimus has shown some promise in improving/stabilizing kidney function in patients with chronic allograft dysfunction, with the caveat that the development of nephrotic or subnephrotic range proteinuria following conversion to sirolimus is relatively common and requires cessation of the treatment. The presence of proteinuria before conversion is predictive of a poor response and should be screened for before considering a switch to sirolimus.[134] Finally, conversion from CNI to sirolimus is associated with regression of Kaposi sarcoma and reduction in the development of squamous cell skin cancers.[135,136] Everolimus is approved for the treatment of advanced renal cell carcinoma.

COSTIMULATORY SIGNAL BLOCKER

Belatacept, given as an intravenous infusion, is composed of human IgG1 Fc fragment linked to the modified extracellular domain of cytotoxic T lymphocyte–associated antigen 4 (CTLA4). It selectively binds B7 (CD80 and CD86) on APCs and prevents their interaction with CD28 on T cells (Signal 2 of T-cell activation). Two studies of living or standard deceased donor recipients (BENEFIT[85]) and those of expanded criteria deceased donor kidneys (BENEFIT-EXT[107]) randomized subjects to higher-dose belatacept, lower-dose belatacept, or CsA, with all receiving IL-2 RA induction, and MMF and steroid maintenance concurrently. Of note, these studies did not compare belatacept to FK, which has been shown to have superior outcomes when compared with CsA. One-year results showed similar graft and patient survival across the groups but significantly higher GFR in belatacept-treated subjects.[137,138] Acute rejection was significantly higher in those treated with belatacept in BENEFIT, but not BENEFIT-EXT.[107,139] However, graft and patient survival were significantly higher at 7 years post transplant in the belatacept- versus CsA-treated group in the BENEFIT Study.[137] An analysis of a subset of patients from these two trials showed a lower rate of development of de novo DSA in belatacept versus CsA at 3 years post transplant.[140]

More recently, several studies have evaluated the safety and efficacy of either de novo belatacept or belatacept conversion, as well its impact on vaccine, infection-related, and malignancy outcomes.[141,142] As in the BENEFIT study, rates of acute rejection were again higher in belatacept-treated groups without an impact on allograft survival or eGFR.[143,144] A 2023 study found that early conversion to belatacept (<6 months post transplant) may improve eGFR compared with later conversion.[145] Patients receiving belatacept may have a blunted response to severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) vaccines and high-CMV risk patients may be higher risk for late CMV infection.[146] The incidence of (mostly central nervous system) posttransplant lymphoproliferative disorder (PTLD) was higher in patients treated with belatacept in both trials and was associated with pretransplant EBV seronegativity. While the current package insert recommends use of belatacept only in those who are EBV seropositive (IgG+), a 2014 analysis of 5 studies and 1535 kidney transplant recipients found no difference in the incidence of PTLD in belatacept versus CNI groups or among those who were EBV seronegative versus seropositive.[147] Data are limited on use in pregnant transplant recipients, though a 2023 case series of 12 recipients with 16 pregnancies reported no birth defects or fetal deaths in 13 live births.[148]

CORTICOSTEROIDS

Corticosteroids have both antiinflammatory and immunosuppressive properties and have played an important role in the immunosuppressive management of transplant recipients for not only induction and maintenance therapy but also treatment of acute rejection. The antiinflammatory effects of steroids are mediated by reduction of proinflammatory molecules including the platelet activated factor (PAF), prostaglandins, leukotrienes, and reduction of the release of tumor necrosis factor-α (TNF-α). Immunosuppressive properties of steroids include prevention of T cell proliferation, inhibition of cytokine production including IL-2, and interference with antigen presentation. Some of these effects are mediated through inhibition of the transcription factor nuclear factor kappa B (NF-κB). Chronic use of steroids potentiates lymphocyte apoptosis, resulting in lymphopenia, and interferes with the leukocyte trafficking.[149]

The combination of these mechanisms make steroids a potent and versatile immunosuppressive drug, though adverse effects related to chronic use are also extensive (e.g., diabetes, hypertension, hyperlipidemia, osteoporosis, avascular necrosis, truncal obesity, hypertrichosis, acne, cataracts) and have led to increasing interest in steroid-sparing regimens. The emergence of new, more potent immunosuppressive drugs has allowed the successful implementation of such protocols (steroid avoidance, minimization, or withdrawal).[150,151] This is reflected by a significant trend toward decreased steroid utilization, with 67% compared with 90% of U.S. kidney transplant recipients maintained on the drug in the years 2021 and 1995, respectively. In 2021, 26% were maintained on tacrolimus/MMF alone. Reported benefits of steroid-sparing strategies are manifold and include improvements in blood pressure, glycemic and lipid control, reduced posttransplant weight gain, better bone mineral density, growth, and physical appearance.[152-155] A Cochrane review of 30 RCTs of steroid avoidance (<2 weeks of exposure) and steroid withdrawal (>2 weeks exposure) demonstrated that such steroid-sparing strategies were not associated with an increase in mortality or all-cause graft loss but were associated with an increase in death-censored graft loss and acute rejection. However, the increased risk of acute rejection was seen in those treated with CsA but not FK-based immunosuppression regimens.[156] Of note, a 2020 randomized study that was halted early found that corticosteroid avoidance in patients initiated on belatacept therapy after rATG induction therapy had increased rates of acute cellular rejection, with or without the addition of FK.[157] Steroid-sparing strategies are a reasonable option for low-immunologic-risk patients who are treated with CNI/MPA-based immunosuppression regimens. Most steroid withdrawal protocols wean steroids within 3 to 6 months of transplant. Late withdrawal from steroids (>1 year post transplant) has been associated with increased acute rejection rates and deterioration in graft function in studies from the CsA/azathioprine era.[158,159] While late withdrawal may be safer today, caution is recommended with this approach.[160]

EVALUATION OF THE RECIPIENT EARLY AFTER TRANSPLANT

DELAYED GRAFT FUNCTION

The lack of improvement of serum creatinine in the immediate postoperative period may signal that there is ischemia-reperfusion injury of the allograft. Such allografts may be categorized as "slow graft function" (SGF) or DGF. SGF does not have a standardized definition and varies based on research study.[161-163] Attempts to classify SGF as prognostic of long-term graft dysfunction are imperfect[164] and hence not typically used in clinical practice.

In contrast, DGF has a standardized definition utilized clinically and in transplant reporting as the need for one or more dialysis treatments within the first week post transplant. Indication for dialysis is not considered; treatment may be for hyperkalemia, volume overload, or underdialysis before the transplant procedure. The designation may theoretically exclude patients with residual native kidney function that undergo preemptive transplant as they may not require dialysis while the allograft is not functioning.[165] The incidence of DGF varies widely depending on transplant center[166] and there may be intracenter and intercenter variation in thresholds for posttransplant dialysis and donor source, with living donors having the lowest rate of DGF (3%)[167] and donation after cardiac death (DCD) kidneys the highest (40.1%). The DGF rate overall in the United States has increased significantly over the past 3 years to 24%, associated with the change from locally prioritized allocation to broader geographic sharing in March 2021 (KAS250) with notable increases in cold ischemia times.[168,169] Ischemic acute tubular necrosis (ATN) is by far the most common cause of DGF.[170]

The impact of DGF is not simply on patient management and decisions for dialysis implementation. It is also due to the negative consequences on graft survival, baseline kidney function, and acute rejection. In a paired analysis of kidneys from DCD donors, those recipients with DGF had a significantly higher adjusted hazard ratio (HR) for overall graft loss at 3 years compared with recipients without DGF (HR 4.31, 95% confidence interval [95% CI], 1.13–16.44). Similarly, analysis of SRTR data from 1997 to 2010 showed DGF was associated with increased risk of graft loss at 1 and 5 years (13.5% and 16.2% increase, respectively, $P < 0.01$) and mortality (7.1% and 11% increase, respectively; $P < 0.001$).[171] There is a dose-dependent impact with higher number of dialysis treatments and longer duration of DGF also associated with worse outcomes.[172-174] Similarly, acute rejection is more frequent as noted in a meta-analysis with a relative risk of acute rejection overall up to 60% higher, depending on duration and severity of DGF.[172,175] Importantly, there are indirect impacts of DGF including reduced probability of having a functioning graft (from 52%–32%), an increased probability of being on dialysis (from 10%–19%), and increased probability of death (from 38%–50%) with a loss of 0.87 quality-adjusted life-years (QALYs) extrapolated over a lifetime.[176]

Recipient-associated risk factors for DGF include male sex, black race, longer dialysis vintage, higher PRA status, and greater degree of HLA mismatching.[170] Donor factors include use of deceased donor kidneys (especially high kidney donor profile index (KDPI) or DCD, discussed later in "Assessing Outcomes in Kidney Transplantation"), older donor age, and longer cold ischemia time. Indeed, a risk calculator has been developed to predict DGF,[177] although its accuracy has been updated.[178] The implementation of ex vivo perfusion of donor grafts has been associated with reduced rates of DGF in the recipient,[179] although pump perfusion is not used consistently across organ procurement organizations.

Diagnosis of DGF includes assessment of clinical, radiologic, and histologic features, although data suggest that biopsy histology may not be as strong a predictor.[180] Oliguria (not anuria) is present for 5 to 10 days or more. Ultrasonography with duplex sonography may demonstrate elevated resistive indices (RI), but this does not discriminate between ATN and rejection and is therefore of limited diagnostic utility.[181] Multiple studies have focused on donor expressed proteins as pretransplant biomarkers of AKI, but these have not been implemented clinically and their utility is uncertain.[182]

Prevention of DGF is important to mitigate the negative outcomes noted earlier. Many centers alter induction and maintenance therapy in the hopes of limiting or ameliorating the propagation of DGF. Use of T cell depleting induction, thought to support CNI minimization, has not been shown to be effective in mitigation of DGF.[42,183] Similarly, while CNI avoidance is commonly used to mitigate the potential vasoconstriction of this agent class on the kidney, the evidence does not show benefit.[184] Finally, an RCT comparing balanced crystalloid solution with saline demonstrated reduced DGF in the balanced crystalloid group (adjusted relative risk 0.74, 95% CI 0.66 to 0.84); $P < 0{\cdot}0001$) with balanced crystalloid.[185] These findings should support improved recipient management in an evidence-based fashion.

Management of the patient during this period is supportive. When early hemodialysis is required, minimal anticoagulation should be used to reduce the risk of postsurgical bleeding. Intradialytic hypotension should also be avoided to prevent further renal ischemic injury. Peritoneal dialysis may be successfully continued post transplant, although it should be avoided if the peritoneum was opened at the time of surgery and may be technically challenging if low-volume, frequent exchanges are required. Drug dosing must be adjusted to level of kidney function and implementation of dialysis. Considerations of other causes of oliguria are noted below. Persistent oliguria may require early biopsy to assess for acute cellular- or antibody-mediated injury. Adjunctive biomarkers such as donor-derived, cell-free DNA are not approved for use in the early post-transplant period. Rejection may be theoretically more common as innate immune responses may activate alloimmune responses in these settings.[170] Therefore a high degree of suspicion for additional complications related to the allograft must be maintained.

EVALUATION AND MANAGEMENT OF ALLOGRAFT DYSFUNCTIN POST TRANSPLANT

MANAGEMENT OF ALLOGRAFT DYSFUNCTION POST TRANSPLANT

Fig. 69.3 describes the causes of allograft dysfunction during the posttransplant period including common timing. Early postsurgical complications are discussed in Table 69.1. As for native renal disease, the initial evaluation of transplant allograft dysfunction should be approached systematically using an anatomic classification of prerenal, intrarenal, and postrenal causes, as described later.

PRERENAL

HYPOVOLEMIA

Hypovolemia may develop after intravascular volume depletion from any cause including gastrointestinal losses. Diarrhea is a common adverse effect of MMF, especially when used with tacrolimus. Similarly, immunosuppression increases the risk of gastrointestinal infections including CMV disease, which can result in diarrhea and volume contraction; more than 50% of kidney transplant patients report experiencing diarrhea.[186] The transplanted kidney is denervated with impaired autoregulation and therefore responds less to changes in circulating blood volume than native kidneys, which can exaggerate prerenal insults.

TRANSPLANT RENAL ARTERY STENOSIS

Transplant renal artery stenosis is the most common transplant vascular complication and is associated with reduced long-term allograft survival,[187,188] occurring most commonly between 3 months and 2 years post transplant.[189,190] The reported incidence varies widely on account of heterogenous diagnostic criteria.[188] Transplant renal artery stenosis may be a consequence of inadequate arterial anastomosis, arterial kinking, or preexisting vascular disease of the donor or recipient.[190] Immune-mediated or infection-related damage to the transplant renal artery also plays an important role in some patients; new posttransplant DSAs and CMV infection are both associated with the development of transplant renal artery stenosis.[191,192] Stenosis may occur in the donor or recipient artery or at the anastomotic site. Resultant renal hypoperfusion activates the renin-angiotensin-aldosterone system and may manifest as worsening or refractory hypertension, fluid retention, flash pulmonary edema, erythrocytosis, or allograft dysfunction.[188,190,193,194] Clinical examination may reveal a new vascular bruit over the graft. Ultrasound

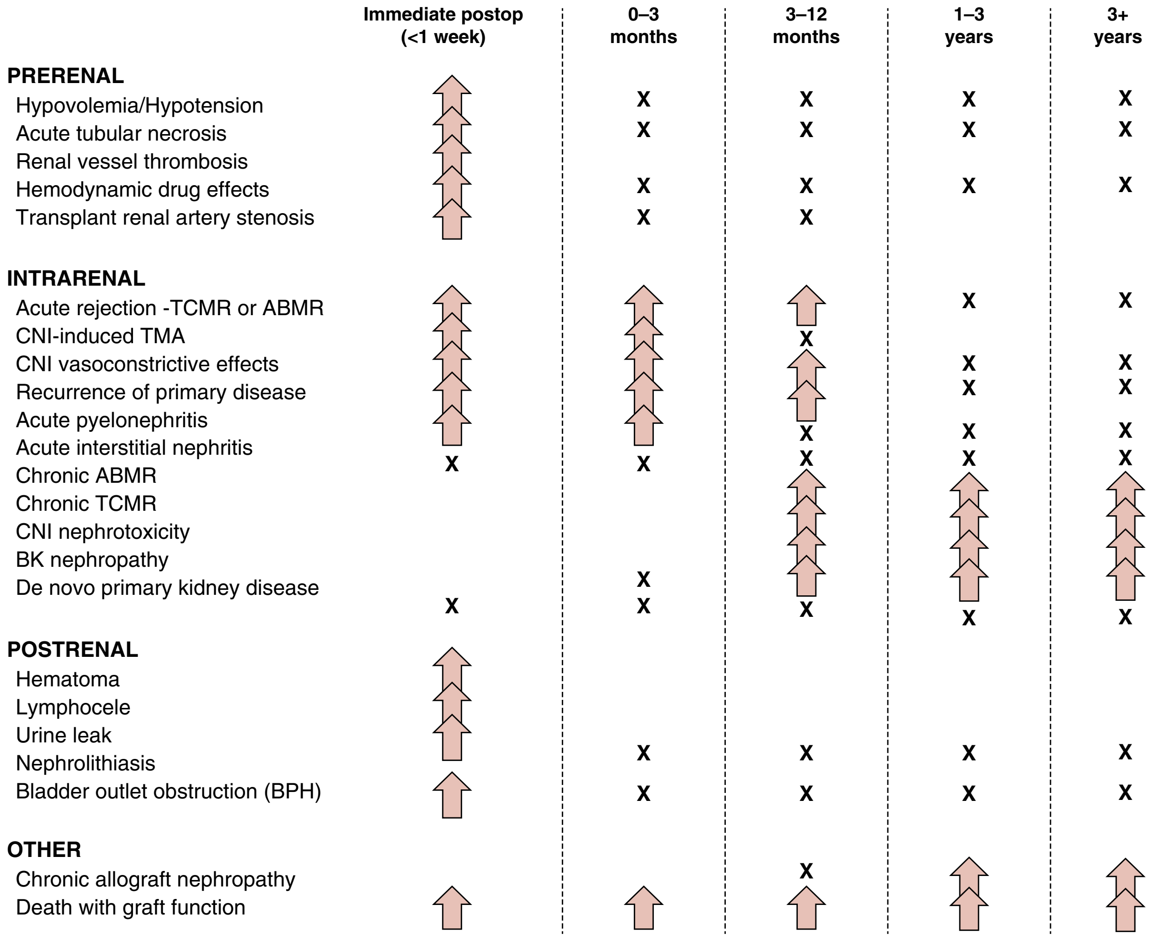

Fig. 69.3 Causes of allograft dysfunction during the posttransplant period. *Arrow,* Increased incidence; *X, occurs,* though less frequently.

with Doppler, magnetic resonance, and computed tomography angiography can support a diagnosis; however, direct angiography is usually required for confirmation, acknowledging the associated risk of contrast-mediated acute kidney injury given the large iodinated contrast load with invasive angiography. CO_2 or minimal contrast angiography can be used to reduce the risk of radiocontrast injury.[195] In the absence of hemodynamically significant stenosis on imaging or progressive kidney dysfunction, transplant renal artery stenosis can be managed conservatively with antihypertensive therapy and consideration of aspirin and/or statin adjuncts as appropriate.[188] For patients with progressive stenosis, worsening kidney function, or uncontrolled hypertension, percutaneous transluminal angioplasty with or without stent placement may be indicated, with early success rates >70% but relatively high risk of recurrence in the following 6 to 8 months.[188,196,197] An open surgical approach can be considered for unsuccessful angioplasty or severe stenosis inaccessible to percutaneous transluminal angioplasty; however, due to potential serious complications, surgery is generally reserved for rescue therapy.[188]

INTRARENAL

Etiologies of intrarenal acute and chronic allograft dysfunction include all causes of native kidney disease (including acute interstitial nephritis, nephrotoxic drugs and radiocontrast, recurrence of the primary disease, de novo disease) in addition to more specific allograft presentations listed as follows.

REJECTION

Hyperacute rejection

With more sensitive techniques for detecting HLA antibodies utilized in the histocompatibility laboratory, hyperacute rejection has become rare, if not absent, in kidney transplantation. However, as some centers transplant incompatible donor/recipient pairs intentionally including those with positive crossmatches (indicating recipient serum containing preformed antidonor HLA antibodies), recognition of this entity still has relevance. Immediately following establishment of vascular anastomoses, circulation of preformed antibody leads to disseminated intravascular coagulopathy, with graft congestion and thrombosis. Histology reveals widespread small vessel endothelial damage, thrombosis, and neutrophil infiltration. This entity may also occur in the setting of ABO-incompatible transplantation, which while performed intentionally in some cases, requires host conditioning. It may also be a consequence in xenotransplantation, when organs of one species are transplanted into another and the host's natural antibodies bind the endothelium on implantation.

Acute rejection

Acute rejection, characterized by a decline in kidney function mediated by a recipient immune reaction against the allograft, may consist of three forms: T-cell mediated rejection (i.e., cellular rejection), antibody (i.e., humoral rejection), or a combination of both. While seen more frequently in the first 6 months of transplantation, the incidence of acute rejection has decreased dramatically in the past 25 years and is now around 7% to 10% in the first 12 months in the United States.[198] It is important to understand that acute rejection may occur at any time post transplant and that the frequency of acute rejection may increase over time,[199] associated with nonadherence with a "mixed" rejection phenotype.[200]

In the current era of immunosuppression, symptoms and signs of acute rejection are often minimal, although low-grade fever, oliguria, and graft tenderness may occur, along with microscopic hematuria and leukocyturia. Detection is primarily based on changes in serum creatinine and, while an insensitive measure, remains the mainstay of detection.

About 25% of transplant centers implement the use of surveillance allograft biopsy to detect the presence of rejection in the absence of graft dysfunction (i.e., subclinical rejection),[201] but this is not uniformly used due to the cost, lack of perceived therapeutic benefit, and patient dissatisfaction and inconvenience.[201] Therefore there is a growing interest in development of early biomarkers of immune system activation (see later). For now, the gold standard for rejection diagnosis is transplant biopsy. Classification is standardized by many centers around the world using the Banff Classification, developed by expert consensus. Histologic findings may be supplemented by gene expression in the biopsy,[202,203] although convincing evidence of clinical utility remains outstanding.[204] Table 69.4 includes the most recent classification schema.[205] Use of artificial intelligence has provided opportunities to improve biopsy classification and diagnostic accuracy based on biopsy descriptions and may prove valuable, particularly in research studies for new treatments.[206]

T cell–mediated rejection

The key histologic features of TCMR include the presence of interstitial inflammation ("i") with mononuclear cells, typically lymphocytes, and the presence of tubulitis ("t"), which refers to infiltration of the renal tubular epithelium by lymphocytes. In more advanced cases, infiltration of mononuclear cells beneath the endothelium or arteritis ("v") may be present. The grade of rejection is based on the extent of these features, although outcomes of treatment are not clearly dependent on grade.

Borderline rejection (BR) remains in the category of TCMR, with Banff 2019 using threshold values of interstitial inflammation involving 10% to 25% of nonsclerotic cortex (i1) with at least mild tubulitis ($t \geq 1$), as isolated tubulitis has not been shown to have a negative effect on transplant outcomes.[207] Some studies have associated BR with significantly higher incidence of subsequent TCMR, interstitial fibrosis and tubular atrophy (IF/TA, i.e., chronicity), and development of de novo DSA.[208-210] Thus BR is indicative of activation of the host immune response.[211]

Finally, it is important to recognize that interstitial inflammation and tubulitis are also seen with BK nephropathy (see later) and need to be distinguished from TCMR using immunostaining for virus, as well as assessment for the presence of viral genome in the recipient. Eosinophilic infiltration may reflect allergic interstitial nephritis, and the exact role in cellular rejection remains unclear.[212]

Table 69.4 Updates of 2019 Banff Classification for Antibody-Mediated Rejection; Borderline Changes, T Cell–Mediated Rejection (TCMR), and Polyomavirus Nephropathy

Category 1: Normal biopsy or nonspecific changes
Category 2: Antibody-mediated changes
Active ABMR: all 3 criteria must be met for diagnosis
1. Histologic evidence of acute tissue injury, including 1 or more of the following:
- Microvascular inflammation (g >0 and/or ptc >0), in the absence of recurrent or de novo glomerulonephritis, although in the presence of acute TCMR, borderline infiltrate, or infection, ptc ≥1 alone is not sufficient and g must be ≥1
- Intimal or transmural arteritis (v >0)b
- Acute thrombotic microangiopathy, in the absence of any other cause
- Acute tubular injury, in the absence of any other apparent cause
2. Evidence of current/recent antibody interaction with vascular endothelium, including 1 or more of the following:
- Linear C4d staining in peritubular capillaries or medullary vasa recta (C4d2 or C4d3 by IF on frozen sections, or C4d >0 by IHC on paraffin sections)
- At least moderate microvascular inflammation ([g + ptc] ≥2) in the absence of recurrent or de novo glomerulonephritis, although in the presence of acute TCMR, borderline infiltrate, or infection, ptc ≥2 alone is not sufficient and g must be ≥1
- Increased expression of gene transcripts/classifiers in the biopsy tissue strongly associated with ABMR, if thoroughly validated
3. Serologic evidence of circulating donor-specific antibodies (DSA to HLA or other antigens). C4d staining or expression of validated transcripts/classifiers as noted above in criterion 2 may substitute for DSA; however, thorough DSA testing, including testing for non-HLA antibodies if HLA antibody testing is negative, is strongly advised whenever criteria 1 and 2 are met

Chronic active ABMR; all 3 criteria must be met for diagnosis
1. Morphologic evidence of chronic tissue injury, including 1 or more of the following:

Transplant glomerulopathy (eg >0) if no evidence of chronic TMA or chronic recurrent/de novo glomerulonephritis; includes changes evident by electron microscopy (EM) alone (cg1a)
Severe peritubular capillary basement membrane multilayering (ptcml1; requires EM)
Arterial intimal fibrosis of new onset, excluding other causes; leukocytes within the sclerotic intima favor chronic ABMR if there is no prior history of TCMR, but are not required

2. Identical to criterion 2 for active ABMR, above
3. Identical to criterion 3 for active ABMR, above, including strong recommendation for DSA testing whenever criteria 1 and 2 are met. **Biopsies meeting criterion 1 but not criterion 2 with current or prior evidence of DSA (posttransplant) may be stated as showing chronic ABMR; however, remote DSA should not be considered for diagnosis of chronic active or active ABMR**

Chronic (inactive) ABMR
1. cg >0 and/or severe ptcml (ptcml1)

C4d staining without evidence of rejection; all 4 features must be present for diagnosis
1. Linear C4d staining in peritubular capillaries (C4d2 or C4d3 by IF on frozen sections, or C4d >0 by IHC on paraffin sections)
2. Criterion 1 for active or chronic active ABMR not met
3. No molecular evidence for ABMR as in criterion 2 for active and chronic active ABMR
4. No acute or chronic active TCMR. or borderline changes

Category 3: Borderline (suspicious) for acute TCMR
Foci of tubulitis (tl. t2. or t3) with **mild interstitial inflammation (i1)**, or mild (t1) tubulitis with moderate-severe interstitial inflammation (i2 or i3)
No intimal or transmural arteritis (v = 0)
Category 4: TCMR
Acute TCMR
Grade IA: Interstitial inflammation involving >25% of nonsclerotic cortical parenchyma (i2 or i3) with moderate tubulitis (t2) involving 1 or more tubules, not including tubules that are severely atrophied
Grade IB: Interstitial inflammation involving >25% of non-sclerotic cortical parenchyma (i2 or i3) with severe tubulitis (t3) involving 1 or more tubules, not including tubules that are severely atrophic
Grade IIA: Mild to moderate intimal arteritis (v1), with or without interstitial inflammation and/or tubulitis
Grade IIB: Severe intimal arteritis (v2), with or without interstitial inflammation and/or tubulitis
Grade III: Transmural arteritis and/or arterial fibrinoid necrosis involving medial smooth muscle with accompanying mononudear cell intimal arteritis (v3), with or without interstitial inflammation and/or tubulitis
Chronic active TCMRe
Grade IA: Interstitial inflammation involving >25% of sclerotic cortical parenchyma (i-IFTA2 or i-IFTA3) AND >25% of total cortical parenchyma (ti2 or ti3) with moderate tubulitis (t2 or **t-IFTA2**) involving 1 or more tubules, not including severely atrophic tubules: other known causes of i-IFTA should be ruled out
Grade IB: Interstitial inflammation involving >25% of sclerotic cortical parenchyma (i-IFTA2 or i-IFTA3) AND >25% of total cortical parenchyma (ti2 or ti3) with severe tubulitis (t3 or **t-IFTA3**) involving 1 or more tubules, not including severely atrophic tubules; other known causes of i-IFTA should be ruled out
Grade II: Chronic allograft arteriopathy (arterial intimal fibrosis with mononuclear cell inflammation in fibrosis and formation of neointima). This may also be a manifestation of chronic active or chronic ABMR or mixed ABMR/TCMR
Category 5: polyomavirus nephropathyf
PVN Class 1
pvl 1 and ci 0-1
PVN Class 2

[a]All updates in boldface type.
From Loupy A, Haas M, Roufosse C, et al. The Banff 2019 Kidney Meeting Report (l): Updates on and clarification of criteria for T cell- and antibody-mediated rejection. *Am J Transplant*. 2020;20(9):2318–2331.

Treatment of T cell–mediated rejection

Treatment of TCMR of grades I and higher is nearly uniformly considered, but the treatment itself has not been standardized. This includes choice of agents, length of treatment, dosing, and follow up.[198] Surveys in both the United States[213] and abroad indicate that intravenous corticosteroids consisting of methylprednisolone 250 to 500 mg given daily for 3 to 5 doses are commonly used. Management includes continuation of maintenance therapy often augmented by higher maintenance doses and more frequent monitoring of serum creatinine. Successful completion of therapy remains debated, whether based on allograft function or a follow-up biopsy,[214] and this has importance for consideration of additional treatment. Lack of return to baseline function (if known) and/or persistent inflammation if biopsy is undertaken supports use of additional therapy. Here, lymphocyte-depleting antibodies are highly effective in treating first rejection episodes. Because of toxicity and cost, these agents are usually reserved for higher-grade rejections based on local practices, as well as in steroid resistance, but the impact of treatments in terms of preventing later rejection or facilitating functional improvements in resistant rejections remains unclear.[215]

Studies have indicated that BR included as part of TCMR episodes contributes to later episodes of TCMR and death-censored graft loss,[199] as well as development of de novo anti-HLA DSA.[209] With regards to BR, treatment is debated and subject to center and individual practices, as well as clinical context of the findings, as summarized by Filippone and Farber.[216]

Implications of T cell–mediated rejection

Biopsy-proven acute rejection remains a primary endpoint for clinical trials for new drug development. However, multiple studies in the modern immunosuppressive era have identified a dissociation with long-term outcomes,[217] albeit recognizing negative impacts of single episodes as noted earlier,[199] as well as the potential negative consequences of antirejection therapy on the recipient.[218] Importantly, response to treatment is a key indicator of long-term impact and consideration of optimal patient management should include consideration of follow-up biopsy and monitoring of eGFR, DSAs, and proteinuria.[219] Use of multiparameter assessments can provide more personalized risk assessment for individual patients.

As already noted, surveillance biopsy is employed at some centers as a standard of care,[213] and numerous studies have identified subclinical rejection as having negative impacts on subsequent graft function and survival[199,210,220] including those with BR.[221,222] Treatment of subclinical rejection remains inconsistent, in part due to lack of data and consistency of approaches.[216]

Chronic active T cell–mediated rejection

Chronic active T cell–mediated rejection (CA-TCMR) was considered a lesion of chronic vasculopathy with inflammation, but this classification schema was changed significantly in 2017 and updated in 2019, with the accumulation of data that inflammation in areas of interstitial fibrosis and tubular atrophy (i-IFTA) were important pathologic features, previously not included in prior Banff iterations regarding rejection.[205] This histologic entity has been associated with late graft failure in multiple studies.[223,224] Importantly, i-IFTA alone is insufficient to make a diagnosis of CA-TCMR, which requires additional criteria, including at least a moderate degree of total cortical inflammation and moderate tubulitis involving cortical tubules other than those that are severely atrophic. Subsequent studies identified TCMR as a risk factor for CA-TCMR.[225,226] Therapy for now remains empiric as there are no large-scale studies or recommendations for treatment but recognition of the significance in portending subsequent graft loss.

Active antibody-mediated rejection type I and type 2

With the introduction of Banff criteria for active antibody-mediated rejection (ABMR) in 2001, there has been a growing understanding about the early recognition and interventions of this entity and its implications for graft dysfunction and graft failure. ABMR is increasingly recognized as a cause of allograft dysfunction. The reported incidence varies from 3% to 12% in the first year after transplant, more frequent in the context of HLA-incompatible kidney transplantation and with the use of surveillance biopsies.[227]

Diagnosis of active ABMR requires the presence of all of the following criteria, developed and updated in 2019 (see Table 69.4) to add diagnostic sensitivity and predictive value for graft outcomes: 1. histologic evidence of acute tissue injury such as microvascular inflammation, intimal or transmural arteritis, acute thrombotic microangiopathy or acute tubular necrosis in the absence of any other apparent cause; 2. evidence of recent antibody-endothelial interaction, usually identified using C4d staining of peritubular capillaries, moderate microvascular inflammation, or increased expression of gene transcripts/classifiers in the biopsy tissue; and 3. serologic evidence of antibody against donor HLA or other antigens, although C4d staining or expression of validated transcripts/classifiers as noted earlier in criterion 2 may substitute for DSA. Importantly, thorough DSA testing, including testing for non-HLA antibodies if HLA antibody testing is negative, is strongly advised whenever criteria 1 and 2 are met. Active ABMR typically occurs early after transplantation but can also occur late, especially in the setting of reduced immunosuppression or noncompliance. Active ABMR may occur alone or with TCMR. In addition, subclinical active ABMR is commonly present on surveillance biopsies of immunologically high-risk kidney transplant recipients and is associated with poor graft outcome.

Active ABMR consists of two clinical phenotypes: Type 1 ABMR results from persistence and/or rebound of preexisting DSAs in sensitized patients and usually occurs early post transplantation. Type 2 ABMR is associated with de novo DSAs and usually occurs a year or more post transplantation. While histologically identical, their clinical presentations differ with type 2 occurring later post transplant, often with class II antibodies.[228] These clinical differences have resulted in varying approaches to management with consensus that has evolved based on expert practice and clinical reports.[229] In the former, plasmapheresis, IVIG, and corticosteroids (grade 1C) are the typical mainstay of therapy while adjuncts include use of complement inhibitor anti-C5 and rituximab (2B), as well as splenectomy in severe cases (3C). In late presentations with the development of de novo DSA, there is no clear-cut effective therapy. Optimization of baseline immunosuppression (1C) and managing nonadherence, if present, are key and may be coupled with specific therapies such as plasmapheresis and IVIG and/or rituximab (3C). However, treatment considerations in this situation may depend on level of baseline graft function, extent of fibrosis and atrophy, and potential negative impacts of more intensified immunosuppression. There are no U.S. Food and Drug Administration–approved therapies for ABMR at the present time, but a number of agents are undergoing randomized trials.[230] Recognizing that de novo DSA is an important mediator of late ABMR and also a negative prognostic factor, there are continued efforts to develop monitoring protocols post transplant based on immunologic risk.[231,232]

Non-HLA antibodies have been identified in preclinical models of graft injury and failure. They have similarly been noted in human transplant recipients and are associated with organ injury and graft failure.[233] However, it has been difficult to link them mechanistically to graft failure.[234] Regardless, it continues to be recommended that when histologic features of ABMR are identified, a thorough assessment of non-HLA antibodies be performed if HLA DSAs are not detectable.[205]

Chronic active antibody-mediated rejection

The reported incidence of CA-ABMR ranges from 4.6% to 20.2% over 1 to 10 years,[227] so it is not an uncommon finding. The Banff 2005 update added the term "chronic active antibody-mediated rejection," currently requiring all three of the following: 1. morphologic evidence of chronic tissue injury evidenced by transplant glomerulopathy, peritubular capillary basement membrane multilayering (on EM), and/or arterial intimal fibrosis of new onset; 2. evidence of recent endothelial-antibody interaction (peritubular capillary C4d staining); and 3. evidence of DSA.[235] Typically, CA-ABMR is clinically characterized by a decline in kidney function, hypertension, and proteinuria. The latter is a strong prognostic factor for graft failure.

Risk factors for developing CA-ABMR include sensitization before transplant including prior transfusions, pregnancy, and transplants; the persistence of preformed antibody in those with incompatible transplants; and the development of de novo DSA, which may be seen with HLA class II mismatch, younger patients, prior TCMR episodes, under immunosuppression, and nonadherence. The role of non-HLA antibodies remains uncertain,[236] but there is growing evidence that innate immune activation is associated with CA-ABMR.[237]

Management may include use of RAAS blockade as tolerated. From an immunologic perspective, there is no known effective therapy and recommendations for management include IVIG, although data are limited.[229] Newer approaches focus on innate immune responses including anti-IL-6 and other therapies targeting antibody production, such as an ongoing international RCT that is under way to assess blockade of the IL-6 receptor[238] and other planned trials using novel approaches.[239]

CALCINEURIN INHIBITOR NEPHROTOXICITY

Acute CNI toxicity is discussed earlier. Chronic CNI exposure has been shown to irreversibly impact all three renal compartments including the vessels, tubulointerstitium, and glomeruli. Histologically, this may present as arteriolar hyalinosis, tubular atrophy, and striped interstitial fibrosis or thickening of the Bowman capsule with or without corresponding focal or global glomerulosclerosis.[99,235] In a study of 120 diabetic kidney transplant recipients (119 of whom underwent SPK) with sequential kidney transplant biopsies, histologic evidence of CNI toxicity defined as "striped cortical fibrosis or new-onset arteriolar hyalinosis and tubular microcalcification" was demonstrated in >50% of biopsies at 5 years and 100% of biopsies at 10 years post transplant.[96] This study did not include a control arm however, and newer studies have suggested a much lower prevalence of interstitial fibrosis at 5 years post transplant, questioning the specificity of arteriolar hyalinosis for the diagnosis of CNI toxicity.[240,241] This, combined with increased evidence for immune-mediated injury in many cases of late allograft failure, has led many to deemphasize the importance of chronic CNI toxicity. Currently, evidence to support CNI withdrawal, conversion, or minimization is lacking in light of the competing risk of allograft rejection. Calcium channel antagonists may be protective against chronic CNI toxicity due to their vasodilatory mechanism preventing the corresponding fall in renal plasma flow and GFR observed with CNI therapy.[241]

HUMAN POLYOMAVIRUS INFECTION AND BK NEPHROPATHY

BK virus (BKV) is a nearly ubiquitous nonenveloped polyoma virus that typically presents as a mild self-limited respiratory event in healthy individuals (usually in childhood), with BKV seropositivity >80% to 90% in people older than 10 years of age.[243,244] Following primary infection, the virus remains latent in renal tubular and uroepithelial cells, with viral reactivation associated with later immunosuppression. This may manifest as isolated viruria or viremia, tubulointerstitial nephritis with slowly progressive graft dysfunction (BK nephropathy), sterile pyuria, hemorrhagic or nonhemorrhagic cystitis, ureteritis, or ureteric stenosis with resultant obstruction. While controversial, emerging data also suggest potential oncogenic properties of BKV.[244] Risk factors for BKV include greater immunosuppression exposure (e.g., thymoglobulin), rejection or ischemia episodes, DGF, decreased cellular immunity, male sex, African American race, ureteral stent placement, and older age.[244]

BK nephropathy is most common in the first 2 years post transplant or following treatment for rejection. Approximately 30% to 40% of renal transplant recipients develop BK viruria, 10% to 20% develop BK viremia, and 1% to 10% progress to BK nephropathy.[244,245] BK viruria and viremia almost always precede BK nephropathy, so urine or plasma BK viral screening offer an opportunity to identify early preclinical viral replication and institute preemptive management strategies to help mitigate progression to BK nephropathy. Given the higher positive predictive value with plasma BK levels, this is the preferred BK screening method utilized by most centers[246]; however, approaches to screening vary and are influenced by local prevalence and economic factors.[247]

The risk of BK nephropathy increases with higher BK viral titers; a plasma BK viral load ≥10,000 copies/mL is generally considered a "presumptive" diagnosis of BK nephropathy. However, given the potential for concurrent histologic diagnoses (including rejection), transplant biopsy is still warranted in most cases. The presence of intranuclear tubule cell inclusions by light microscopy should raise suspicion; diagnosis is confirmed by immunohistochemistry using antibodies against BK viral proteins (SV-40). However, BKV is associated with focal disease primarily effecting the renal medulla, with histologic evidence of BK nephropathy missed in ~30% of cases. A negative biopsy therefore does not rule out BK nephropathy, and with high clinical suspicion, a nonconfirmatory biopsy may need to be repeated.[243]

The mainstay of BKV treatment is reduction in immunosuppression.[248] A number of strategies have been proposed, but most commonly, significant viremia or biopsy evidence of nephropathy will prompt dose reduction or discontinuation of the antiproliferative agent (usually MMF). If this intervention fails to result in a favorable viral response, the dose of CNI is reduced by 30% to 50% with continued monitoring of viral titers and for the precipitation of concurrent rejection in light of reduced immunosuppression. Switching tacrolimus to cyclosporine may be considered; some, but not all RCTs, have shown less BKV in patients treated with cyclosporine than tacrolimus,[249-252] whereas most studies show benefit with mTORi versus mycophenolate.[253-255] Tacrolimus is believed to increase in vitro BKV replication independent of its immunosuppressive properties, whereas cyclosporine and mTOR inhibitors appear to directly inhibit BKV replication (although this may simply represent a de facto reduction in immunosuppression).

A number of adjunct strategies may be considered in subjects who either fail to respond to immunosuppression reduction or in whom very aggressive immunosuppression reduction is unattractive because of their high-risk immune status; however, evidence of benefit is conflicting. IVIG preparations (which contain high titers of neutralizing antibodies against all BKV genotypes) have also shown promise in treating BKV that has not responded to immunosuppression reduction in isolation.[256] IVIG antiviral benefit has been demonstrated fairly consistently and is seemingly well tolerated in small case series, yet large trial data are lacking and cost has limited enthusiasm at many centers, pending more comprehensive and robust data demonstrating efficacy.[244] A number of small series have suggested that treatment with leflunomide (usually as a substitute for an antimetabolite) may result in enhanced BK viral clearance, although in the absence of clinical trials its efficacy remains contentious with no clear benefit demonstrated in meta-analyses.[257-259] Similarly, the antiviral cidofovir has been associated with a reduction in BK viral load in some, but not all, case series with significant potential for nephrotoxicity and no antiviral benefit demonstrated in larger meta-analyses.[259,260] Neither leflunomide nor cidofovir is currently recommended for the treatment of BK viremia or nephropathy. Finally, hopes for fluoroquinolones as a therapy for BKV were dashed by two randomized controlled trials that showed levofloxacin to be no better than placebo at 1. treating or 2. preventing BK viremia/viruria in kidney transplant recipients.[261,262]

ACUTE GRAFT PYELONEPHRITIS

Urinary tract infections (UTIs) are relatively common in kidney transplant recipients, with an estimated 10% to 20% of patients experiencing acute graft pyelonephritis (AGPN) posttransplant. UTIs and AGPN may occur at any period but are most frequent in the first 3 to 6 months after transplantation because of catheterization, stenting, and aggressive immunosuppression.[263] In addition to standard risk factors including anatomic abnormalities and neurogenic bladder, other risk factors for UTIs in transplant recipients include shorter ureters and increased urinary reflux due to impaired antireflux mechanisms at the vesicourethral anastomosis. Fever, allograft pain, and leukocytosis are usually more pronounced in acute pyelonephritis than in rejection. Diagnosis requires urine culture, but empiric antibiotic treatment should be started immediately if there is clinical suspicion. Delay in treatment can lead to rapid clinical decline in the immunosuppressed patient. The most commonly implicated microorganisms are gram-negative bacilli, coagulase-negative staphylococci, and enterococci. Kidney function usually returns to baseline quickly with antimicrobial therapy and volume expansion. Recurrent pyelonephritis requires investigation to exclude underlying predisposing urologic abnormalities.

ACUTE THROMBOTIC MICROANGIOPATHY

TMA after kidney transplantation is a rare but serious complication[264] that can present as either a de novo process (often in the setting of preexisting genetic susceptibility) or a recurrence of a primary disease, typically related to dysregulated overactivation of the alternative complement pathway (e.g., in atypical hemolytic uremic syndrome [aHUS]).[265] In the setting of an acquired or genetic predisposition, triggers of new TMA after transplant include CNIs,[266] ABMR,[267] viral infections (including CMV and BKV), ischemia-reperfusion injury, and the recurrence of a previously undiagnosed primary disease. Patients with complement-mediated TMA as a cause for ESKD have a high risk of recurrence in the transplant allograft, which, without prophylactic therapy, ranges from 50% to 100% depending on the underlying complement abnormality; complement factor (CF) I, CFH, CFB, and C3 mutations portend the highest risk of posttransplant recurrence.[265]

Posttransplant TMA occurs most commonly in the first 3 months after transplant but can occur at any time. It is often accompanied by increasing plasma creatinine and lactate dehydrogenase levels, thrombocytopenia, falling hemoglobin level, schistocytosis, and low haptoglobin concentrations, but it may also present in a localized fashion with isolated renal insufficiency, proteinuria, or arterial hypertension. A TMA diagnosis can be overlooked because thrombocytopenia and anemia occur commonly after transplantation in the setting of ATG induction therapy and occasionally with MMF maintenance immunosuppression. The diagnosis is confirmed by allograft biopsy, which shows endothelial damage and, in severe cases, thrombosis of glomerular capillaries and arterioles.

Early diagnosis of TMA is essential to salvage kidney function. There are no controlled trials of therapy for de novo TMA after transplant; however, withdrawal of potential offending agents and neutralization of any potential inciting triggers should be a priority. If immunosuppression-related TMA is suspected, an initial measure is to switch CNIs (either from tacrolimus to cyclosporine or cyclosporine to tacrolimus) or discontinue the CNI drug class altogether in favor of a belatacept or mTORi-based immunosuppression regimen. Other potential underlying etiologies should also be sought including ABMR; in ABMR-associated TMA, the mainstay of treatment is plasmapheresis, optimization of immunosuppression, and often IVIG. In patients with de novo TMA that fails to improve with the above measures, plasma exchange has been recommended, yet while plasma exchange has been shown to improve hematologic parameters, its effect on renal outcomes has been inconsistent at best.[268-270] Eculizumab is an anti-C5 monoclonal antibody that inhibits the alternative complement pathway and has been employed successfully for the treatment of posttransplant TMA relating to aHUS recurrence or as a prophylactic therapy in the setting of known high-risk complement mutations.[271,272] However, even in the absence of a known complement mutation or prior diagnosis of aHUS, in many cases, de novo posttransplant TMA relates to complement overactivation, suggesting a role for eculizumab in this population as well, particularly amongst nonresponders to initial strategies.[265,273]

RECURRENCE OF PRIMARY GLOMERULONEPHRITIS

GN is the cause of ESKD in 30% to 50% of patients who undergo subsequent kidney transplant,[274] with an overall recurrence rate of 7% to 55%.[275] The true incidence of recurrence is difficult to estimate given most relevant studies are small and retrospective with variable follow-up periods.[274] Given the heterogenous pathology and acuity of the various GN subtypes, the presentation and consequences of recurrence differ by GN diagnosis. An Australian study of patients with biopsy-proven GN found a 10-year incidence of graft loss from recurrence of 8.4%, rendering recurrent GN the third most common cause of graft loss.[276] Most GN recurrence results in allograft failure 3 to 5 years post transplant; however, certain etiologies may present early and aggressively post transplant (e.g., MPGN, FSGS).[277] Recurrent GN typically presents similarly as for native kidney disease, with decreasing eGFR, proteinuria, or hematuria. Treatment is typically supportive and includes RAAS inhibition and strict blood pressure control. Immunosuppressive therapy for the prevention or treatment of posttransplant GN has shown limited benefit. There is a paucity of data to support specific preventative or treatment strategies for most recurrent GN presentations, with treatment recommendations extrapolated from native kidney disease and outcomes in the transplant setting poorly described. However, targeted therapies have been more widely accepted for some posttransplant GN recurrences including plasmapheresis in patients with recurrent primary FSGS. A summary of recurrent GN presentations and recommendations posttransplant is shown in Table 69.5, acknowledging the inconsistencies in reporting and lack of consensus regarding therapeutics for most conditions.

POSTRENAL

Most urologic complications are secondary to technical factors at the time of transplant and manifest themselves in the early postoperative period, but immunologic factors may play a role in some cases.

URINARY TRACT OBSTRUCTION

Although urinary tract obstruction can occur at any time after transplantation, it is most common within the first 6 months. It can occur at any location but most frequently involves the site of implantation of the ureter into the bladder. Intrinsic causes include ureteric kinking, intraluminal blood clots or slough material, poor implantation of the ureter into the bladder, and fibrosis of the ureter due to ischemia or rejection. As described earlier, BKV can lead to ureteric stenosis, and rarely renal calculi may also cause obstruction. Extrinsic causes include an enlarged prostate in elderly men (causing bladder outlet obstruction) and compression by a lymphocele, urinoma, or other fluid collection. The transplanted allograft is denervated: Urinary tract obstruction is often asymptomatic and should always be considered in the differential diagnosis of allograft dysfunction. This is of particular importance given that both upper and lower obstruction of the solitary graft can precipitate acute and substantial graft dysfunction. Ultrasound often demonstrates hydronephrosis, but some dilation of the transplant urinary collecting system may be normal and is often seen in the early postoperative period due to anastamotic edema and increased urine output. Serial scans showing worsening hydronephrosis may be needed to confirm the diagnosis. Renal scintiscan with diuretic washout is useful in equivocal cases. Percutaneous antegrade pyelography is the best radiologic technique for determining the site of obstruction and can be combined with interventional endourologic techniques. In expert hands, endourologic techniques (e.g., balloon dilation, stenting) may be effective in treating ureteric stenosis and stricture. More complicated cases require open surgical repair. Extrinsic compression requires specific intervention such as draining or fenestration of a culprit fluid collection. Obstruction in the early postoperative period due to an enlarged prostate should be managed with bladder catheter drainage and standard urologic therapies including tamsulosin.

Routine intraoperative ureteric stenting at the time of kidney transplantation has significantly reduced the risk of major urologic complications post transplant (urine leak and/or stenosis decreased from a median of 7.0% in unstented patients to 1.0% in stented patients). When UTI prophylaxis is employed, there is no increased infectious risk in stented versus unstented transplant recipients.[29] Intraoperative ureteric stenting is now considered standard practice in most transplant centers, although there remains some disagreement regarding the optimal timing for stent removal.

LATE ALLOGRAFT DYSFUNCTION

Our understanding of late allograft failure has evolved from a single presentation of "chronic rejection" to the recognition of specific entities that require proactive clinical monitoring and biopsy diagnosis,[278,279] encompassing both

Table 69.5 Recurrent Glomerulonephritis After Kidney Transplantation[274,277,416,417]

Primary Disease	Recurrence Rate	Graft Loss With Recurrence	Presentation	Clinical Predictors of Recurrence	Treatment*
Primary FSGS	20-50% for primary; rare for familial	13-20% at 10 years	Can manifest as massive proteinuria (nephrotic range in 80%) within hours to weeks post transplant, hypertension, graft dysfunction	White ethnicity Younger age Rapidly progressive FSGS in primary course Recurrence (especially early) in a prior transplant Living related donors (controversial)	Preemptive perioperative plasmapheresis for 1-2 weeks pretransplant for patients at high risk Early plasmapheresis for relapse, poor prognosis if delayed Consider high-dose CNIs, ACE inhibitors, high-dose steroids, cyclophosphamide, rituximab
Anti-GBM Disease	Rare if anti-GBM antibody negative x 6 months at transplant	Frequent if recurrence	Acute graft dysfunction, RPGN	Positive anti-GBM antibody within 6 months of transplant (transplant should be delayed) Alport syndrome may precipitate de novo anti-GBM disease post transplant	No established therapy Plasmapheresis, Cyclophosphamide, pulse steroids
IgA nephropathy	13-53%, reflecting varying threshold for biopsy	~10% over 10 years	Microscopic or gross hematuria, proteinuria, slow decline in graft function, RPGN. Typically occurs late post transplant.	Younger age Prior graft loss to recurrence Steroid-free maintenance regimen (controversial) No induction therapy Crescentic/rapidly progressive disease in native kidneys Living related donors (controversial)	No established therapy Consider ACEi or ARB for proteinuria, CNI, steroids Cyclophosphamide or rituximab if crescentic disease recurrence/RPGN Budesonide has not been studied in transplant recipients.
Lupus nephritis	2-9%, histologic recurrence in up to 30%	3% at 5-10 years	Typically recurs in first 10 years post transplant. Microscopic or gross hematuria, proteinuria, slow decline in graft function	Transplant while disease not quiescent; usually 6-12 months of disease quiescence Antiphospholipid syndrome is a risk of thrombosis	No established therapy MMF Anticoagulation if antiphospholipid syndrome before transplant
ANCA vasculitis	0-20%	6-8% at 5-10 years	Microscopic or gross hematuria, proteinuria, graft dysfunction	Transplant while disease not quiescent Positive ANCA serology does not appear to influence relapse.	No established therapy Consider cyclophosphamide, rituximab ± IVIg
MPGN	Immune complex: 20-33% DDD/C3GN: 67-100%	Immune complex: High DDD/C3GN: 34-66%	Proteinuria, hematuria, graft dysfunction. DDD occurs later than C3GN and usually isolated graft dysfunction. Low C3 levels	C3 glomerulopathy subtype Monoclonal gammopathy, lower serum complement levels, higher protein, crescents in native biopsy Rapid progression to ESKD	No established therapy Eculizumab for C3 glomerulopathy-limited success, cyclophosphamide, rituximab Treat underlying monoclonal gammopathy/ infection

Continued on following page

Table 69.5 Recurrent Glomerulonephritis After Kidney Transplantation[274,277,416,417] (Continued)

Primary Disease	Recurrence Rate	Graft Loss With Recurrence	Presentation	Clinical Predictors of Recurrence	Treatment*
Membranous nephropathy	10-30%	10-50% at 10 years	Minimal proteinuria to nephrotic syndrome	Higher APLA2R titers Persistent elevated APLA2R titers	No established therapy Treatment of secondary factors Consider RAAS inhibition, rituximab, CNI, cyclophosphamide

[a]Treatment for most recurrent glomerulonephritis after kidney transplant is derived from studies in native kidney disease and utilized with limited data in transplant populations.

ACE, Angiotensin-converting enzyme; *ANCA,* antineutrophil cytoplasmic antibody; *Anti-GBM,* antiglomerular basement membrane; *APLA2R,* anti-phospholipase 2 receptor antibody; *ARB,* angiotensin receptor blocker; *C3GN,* C3 glomerulonephritis; *CNI,* calcineurin inhibitors; *DDD,* dense deposit disease; *ESKD,* end-stage kidney disease; *FSGS,* focal segmental glomerulosclerosis; *IViG,* intravenous immunoglobulin; *MMF,* mycophenolate mofetil; *MPGN,* membranoproliferative glomerulonephritis; *RAAS,* renin-angiotensin-aldosterone system; *RPGN,* rapidly progressive glomerulonephriti.

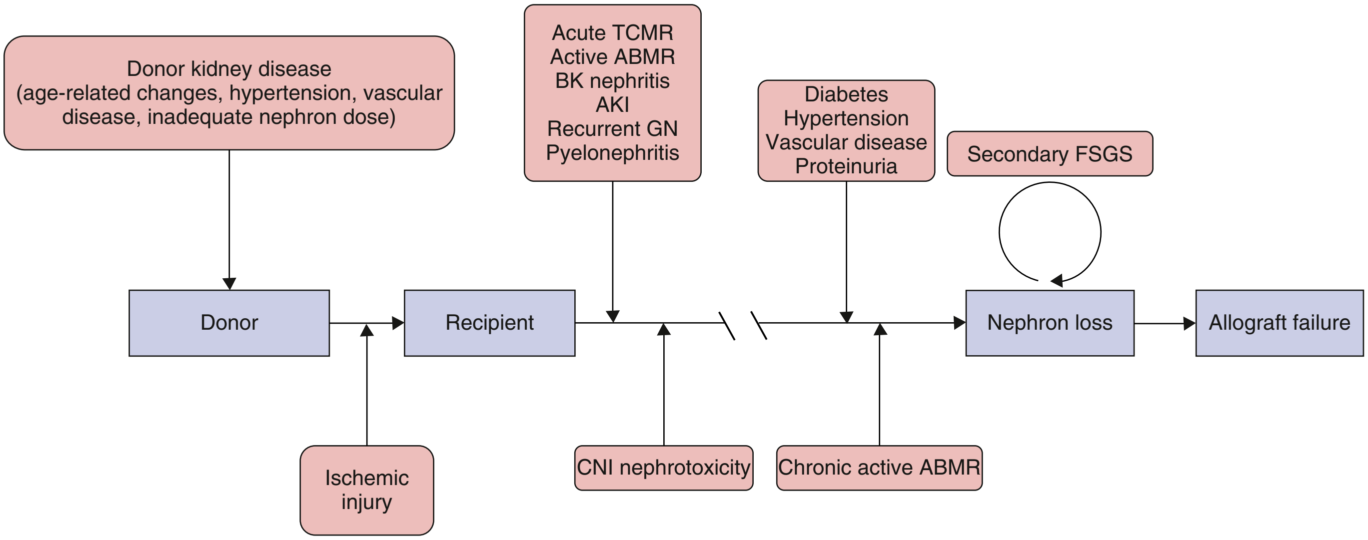

Fig. 69.4 Algorithm for management of persistent delayed graft function. The presence of antidonor human leukocyte antigen (HLA) antibodies should prompt immediate biopsy in this setting.

immune-mediated and non–immune-mediated injuries (Fig. 69.4). As a response to injury, there are the expected tissue remodeling and repair processes. However, if inflammation persists, which is not uncommon in the transplant setting, the resulting maladaptive response is matrix deposition and/or fibrosis. This ultimately leads to declining graft function and finally failure. With our advancing knowledge of the multiple etiologies and mechanisms, enhanced by more recent cohort studies in humans, there is an opportunity to identify those at greater risk to initiate new strategies to ameliorate the process.[280] There has been a renewed focus on management of the patient with late graft failure, regardless of etiology, with emphasis on management of the progressive decline in kidney function and immunosuppressive management. The key here is the determination of access and suitability for repeat transplantation versus dialysis care versus conservative management.[281] Opinions differ on how immunosuppression should be reduced and which agents,[282] dependent in part on the likelihood of repeat transplant in the near term versus maintaining graft function, albeit at a lower but sustainable level.[283,284] Moreover, prior concerns about toxicity of immunosuppression with higher infections and mortality have been challenged on the basis of findings from a prospective cohort of recipients of failed kidney transplants on dialysis demonstrating that maintenance of some immunosuppression was not associated with an increased risk of hospitalization due to infection and, moreover, a lower risk of mortality.[285] Finally, the timing of vascular access and start of renal replacement therapy is another challenge with improvements needed in communications between the transplant center that is frequently managing these patients and their primary nephrologist.[281] Appropriately, shared decision making with the patient is crucial insofar as providing a realistic appraisal of suitability for transplant, complex management of comorbidities, identification of potential donors, and

Table 69.6 Routine Surveillance Laboratory Testing After Transplantation

	<1 Month	1-2 Months	2-6 Months	6-24 Months	>24 Months
Complete blood cell count	Twice weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Basic metabolic panel/glucose/ phosphorus	Twice weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Drug level[a]	Twice weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Liver enzymes	Weekly	Weekly	Every 2 weeks	Monthly	Every 6-12 months
Urinalysis	Twice weekly	Weekly	Every 2 weeks	Monthly	Every 2-3 months
Lipid profile			Annually	Annually	Annually
Urine protein creatinine ratio[b]			3 Monthly	6 Monthly	Annually
BK polymerase chain reaction	Monthly	Monthly	3 Monthly	3/6 Monthly	
Parathyroid hormone[c]			3 Monthly'	6 Monthly'	Annually'
Donor-specific antibodies[d]	[d]	[d]	[d]	[d]	[d]

[a]Depending on the immunosuppressive regimen, to include tacrolimus, cyclosporine, mycophenolic acid.
[b]Frequent UPCR monitoring for patients with end-stage renal disease from focal segmental glomerulosclerosis.
[c]Posttransplant parathyroid hormone monitoring frequency determined by individual parathyroid hormone and calcium levels.
[d]Monitoring frequency determined by patient risk profile.
Standard target levels include:
- CsA (C0) 150-300 ng/mL early and 100-200 ng/mL late.
- CsA (C2) 1400-1800 ng/mL early and 800-1200 ng/mL late.
- Tacrolimus (C0) 8-12 ng/dL early and 5-8 ng/dL late.

understanding the potential immunologic hurdles after a failed transplant.

MONITORING

Signs and symptoms of graft dysfunction often present late, after significant graft injury has already occurred. Screening is required to assess for graft dysfunction, manifestations of drug toxicity, and viral complications and to ensure immunosuppression is at target post transplant. Recommended screening practices and immunosuppression drug target levels are shown in Table 69.6. Protocol posttransplant DSA monitoring is performed at many centers on all or select groups of patients but is certainly warranted in immunologically high-risk recipients and at the time of "for clinical indication" biopsy. DSA may be formed before transplant (preformed) or may occur de novo post transplant. De novo (versus preformed) DSA is associated with worse allograft outcomes.[286] DSA may be accompanied by 1. normal renal function and histology, 2. normal renal function and ABMR histology (subclinical ABMR), 3. overt ABMR, or 4. CA-ABMR. Treatment of DSA depends on accompanying clinical and histologic findings. The detection of an isolated de novo DSA warrants strong consideration for allograft biopsy (irrespective of graft function) and, in some centers, will trigger specific therapies such as IVIG. DSAs may be further risk-stratified by assessing their ability to bind complement (C1q assay) or typing their (IgG) subclass. Treatment of DSA accompanied by ABMR is discussed separately.

Many transplant centers perform elective protocol biopsies at set time points after transplant irrespective of graft function. The merits and demerits of such an approach are keenly debated.[287] To our knowledge, there are no data showing that widespread screening with protocol biopsies in the tacrolimus-MMF era results in better outcomes.[288] Protocol biopsies may, however, have a useful role in selected high immunological risk groups.[289]

There are two major weaknesses in transplant rejection monitoring and diagnosis in its current form: 1. Graft dysfunction (as evidenced by creatinine rise) may occur relatively late in a rejection episode, resulting in a delay in diagnosis and treatment, and 2. Kidney biopsy, the gold standard for rejection diagnosis, is expensive, invasive, prone to sampling bias, and risky. The development of biomarkers that can aid in the early detection of graft dysfunction or even obviate the need for biopsy is therefore highly desirable. Noninvasive methods with potential for allograft surveillance include blood gene expression profiles, mRNA, urinary levels of chemokines, proteomic and metabolic signatures of rejection in the blood or urine, and donor-derived cell free DNA (dd-cfDNA), eTable 69.3. There are strengths and limitations to each method, with inconsistent support and enthusiasm for currently available assays by the transplant community.

dd-cfDNA measurement has been shown to associate with (and predate) clinically significant allograft injury and is thus a potentially useful noninvasive marker of subclinical rejection. The technique utilizes the presence of single nucleotide polymorphisms (SNPs), which are nucleotides

eTable 69.3 Noninvasive Diagnostic Testing for Acute Allograft Rejection in Kidney Transplant Recipients

- Donor-specific antibodies (DSA)
- Donor-derived cell-free DNA (dd-cfDNA)
- Blood gene expression profiles (Trugraf)
- Urinary mRNA
- Urinary levels of chemokine
- Proteomic and peptidomic signatures of acute rejection in urine and blood samples
- IFN-γ enzyme-linked immunosorbent spot (ELISPOT) assay

From Thongprayoon C, Vaitla P, Craici IM, et al. The use of donor-derived cell-free DNA for assessment of allograft rejection and injury status. *J Clin Med*. 2020;9(5):1480. Published 2020 May 14.

at specific positions in the human genome that commonly vary among individuals.[290,291] The most commonly used assay analyzes >250 different SNPs, allowing the test to distinguish donor from recipient cfDNA without a donor blood sample. The assay reports dd-cfDNA as a percentage of total cfDNA in the recipient's blood. A dd-cfDNA level >0.5% to 1% likely reflects underlying graft injury. The test appears sensitive for microvascular inflammation and ultimately may prove useful in identifying those patients with subclinical ABMR,[292] but it cannot be reliably used in those patients with prior kidney transplants or other solid organ transplants. Furthermore, the utility of predicting impending graft compromise in low-immunologic-risk recipients has been questioned.

TruGraf (Eurofins Transplant Genomics; Framingham, Massachusetts) is a DNA microarray-based gene expression blood test for subclinical rejection surveillance in patients with stable allograft function that appears to outperform dd-cfDNA in identifying cellular, but not antibody-mediated rejection episodes.[293] The use of molecular diagnostic tools, such as the "molecular microscope," which employs a microarray-based approach to measure differential gene expression in renal biopsy tissue, have promise and may prove particularly useful in the diagnosis and prognosis of ABMR.[294]

OUTCOMES IN KIDNEY TRANSPLANTATION

SURVIVAL BENEFITS OF KIDNEY TRANSPLANTATION

Comparison of survival between the general dialysis population and transplanted patients is greatly affected by selection bias, as only relatively healthy patients are referred (and listed) for transplantation. Thus comparisons among patients on the waiting list who do or do not receive a transplant are usually performed instead. Of course, such analyses assume that the two groups (those who have undergone transplant surgery and those still on the list) can otherwise be matched, which is not necessarily true.

One meta-analysis showed an overall survival benefit of 55% (adjusted HR 0.45, 95% CI 0.39–0.54) for transplantation versus remaining on the waitlist.[295] There was significant heterogeneity across subgroups, however, with the greatest benefit in studies from Oceania. When stratifying by patient risk, even high-cardiovascular-risk recipients of expanded criteria donors achieved an equal risk to waitlisted dialysis patients by 180 days after transplant and a survival benefit after 521 days.[296]

SHORT- AND LONG-TERM OUTCOMES IN KIDNEY TRANSPLANTATION

The acute rejection rate has fallen substantially over the past 25 years. With the development of immunosuppression regimens that target early rejection, the rate of acute rejection in the first year post transplant has improved over time and is currently around 7% to 10%.[297] Short-term graft survival from all causes has improved. In the United States, 1-year graft survival was 87.7% for deceased and 93.9% for living donors between 1995 and 1999 and increased to 94.3% and 97.8%, respectively, in 2014–2017.[298] There has also been an improvement in long-term allograft survival. This improvement is seen most prominently in the recipients of living kidney donor transplants.[30] Median graft survival for deceased and living donor kidneys was 8.2 and 12.1 years in 1995–1999; this increased to 11.7 and 19.2 years, respectively, in 2014–2017. This reflects an increase in graft survival of 43% for deceased and 59% for living donors over a 20-year period.[298]

Beyond the first posttransplant year, the principal causes of kidney allograft loss are death with a functioning graft and chronic allograft injury (CAI); less common causes are late acute rejection and recurrent disease.[240] The primary causes of death remain cardiovascular disease, infection, and malignancy (Fig. 69.5).[216,217,299] In children, death is a much less common cause of allograft loss; conversely, in the elderly, death is more common.

FACTORS AFFECTING KIDNEY ALLOGRAFT SURVIVAL

Prospective studies and analyses of registry data have shown that many variables influence kidney allograft survival. These can be considered as donor, recipient, donor-recipient pairing, or surgical factors, listed in Table 69.7. Many of them contribute to the development of chronic allograft injury and have been discussed earlier. The roles of the KDPI and DCD donors are discussed as follows.

KIDNEY DONOR PROFILE INDEX

The KDPI is a measure of donor kidney function calculated from the kidney donor risk index (KDRI), which is based on 10 donor characteristics and is a modified version of the predictive tool first described by Rao and colleagues.[300] The KDPI is determined relative to all deceased donor kidneys recovered in the United States in the preceding year and is expressed as a percentile score with 0% and 100% signifying excellent quality and marginal organs, respectively. For example, a KDPI of 70% would indicate a graft failure risk that is higher than 70% of kidneys recovered in the United States the prior calendar year.[301]

DONATION AFTER CARDIAC DEATH DONORS

There has been a significant increase in the use of DCD kidneys. [302] DCD donors can be sub-classified as "uncontrolled" or "controlled." Uncontrolled donors are either unsuccessfully resuscitated or present dead on arrival to hospital, while controlled donors suffer a cardiac arrest following the withdrawal of life support in the intensive care unit or operating room immediately before donation. The duration of warm ischemia time is likely to be significantly greater in the setting of uncontrolled donation.[303] Protocols for managing DCD kidneys vary from center to center. In uncontrolled donation, isolated perfusion of the kidneys with cold preservation solution can be achieved using double-balloon aortic catheterization (with balloons inflated in the aorta above and below the renal arteries) to minimize warm ischemia time.

Short-term outcomes (such as rates of DGF and primary nonfunction) are inferior to those seen for neurologic determination of death (NDD) donors; the adjusted risk

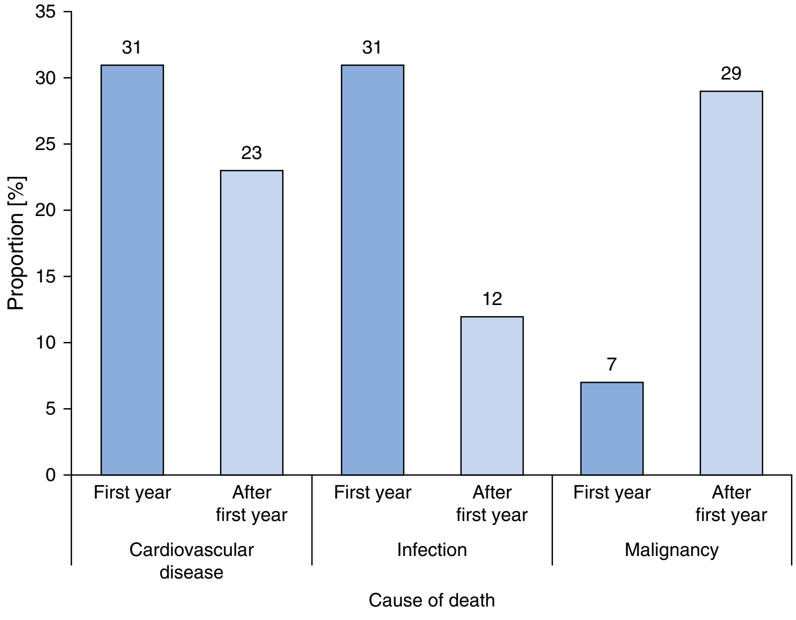

Fig. 69.5 Principal causes of death after kidney transplantation in patients with functioning kidney allografts.

of DGF is nearly three times higher in DCD than NDD donors.[304] However, long-term outcomes of DCD organs are similar to those from NDD donors, including for more marginal donor kidneys.[304,305]

ALLOCATION

Maximizing the life span of donated organs is a key goal in kidney transplantation. The criteria used for allocation of deceased donor allografts can have an important effect on overall allograft survival. A purely utilitarian approach (to maximize allograft survival) would direct organs only to the youngest and healthiest, maximizing the "life years from transplant" gained from transplantation.[306] In practice, a balance must be struck between utility and equity (ensuring that anyone medically fit for a transplant has a reasonable chance of obtaining one).[307]

ORGAN DISCARD

The number of potential transplant kidneys discarded has been increasing over time. Currently about 25% of deceased donor kidneys recovered for transplant are discarded; this is consistent across all age groups and most KDPI categories.[308] This is despite a growing mismatch between the number of people waiting for a deceased donor kidney and the number of available kidney donors.[309] Significant variability in deceased donor acceptance criteria has been demonstrated between transplant centers[310]; however, factors commonly associated with organ decline include a higher KDPI, older age, female sex, Black race, obesity, diabetes, hypertension, hepatitis C positivity, and "donor quality concerns."[309] Despite donors with AKI resulting in acceptable posttransplant graft outcomes, a terminal creatinine >2 mg/dL is associated with a 7-fold increase in the risk of organ discard.[311] Notably, even patients who receive "marginal" deceased donor kidneys achieve a survival benefit versus remaining on dialysis;[312] the 1-year graft survival rate for recipients of marginal kidneys where the other kidney was discarded exceeds 90%.[309] Likewise, a deceased donor kidney with a KDPI >85% confers a better survival at 5 years than declining that offer in favor of waiting for a lower KDPI option, particularly for older patients with long anticipated wait times.[313]

DEATH WITH GRAFT FUNCTION

The mortality rate among kidney transplant recipients is lower than that for patients with ESKD on dialysis; however, despite advances in immunosuppression and antiviral therapies over time, kidney transplant recipients continue to experience increased mortality compared with the age-, sex-, and race-matched general population.[314] Beyond the first year post transplant, death with graft function is the leading cause of allograft loss.[315] Causes of death with graft function vary geographically and with time since transplant; however, in the United States and Australia/New Zealand, they include cardiovascular events, infection, and malignancy.[315,316]

CARDIOVASCULAR DISEASE

Although the relative risk of cardiovascular death has improved over time, cardiovascular disease remains a leading cause of death in kidney transplant recipients.[316,317] Despite pretransplant screening, the cumulative incidence of myocardial infarction (MI), stroke, and de novo peripheral arterial disease are 11%, 7%, and 24%, respectively,[318-320] with some studies suggesting that transplant recipients may experience cardiovascular events with an incidence of 3.5% to 5% per year; 10 to 50× that observed in the general population.[317,321] De novo congestive heart failure is also common.[322] The kidney transplant recipient population is enriched with traditional cardiovascular risk factors such as smoking, diabetes mellitus, obesity, preexisting cardiovascular disease, and hypertension. However, nontraditional CKD-related and transplant-associated risk factors are also important and include acute rejection, chronic inflammation, immunosuppression side effects, and DGF.[323,324] Reduced allograft function is an important and underrecognized risk for cardiovascular morbidity and mortality (15% increased CVD mortality for every

Table 69.7 Donor, Recipient, Donor-Recipient Paring, and Surgical Factors Associated With Outcomes After Kidney Transplantation

Donor Factors	
Living donor	Living donor allografts are superior to deceased donor allografts. Likely reflects: Very healthy living donors, the absence of brain death, benefits of elective vs. semiurgent surgery, minimization of ischemia-reperfusion injury, higher nephron mass.
DCD	Short-term outcomes (e.g., DGF, primary nonfunction) with DCD donors are inferior to those with NDD donors. Long-term outcomes of DCD organs (from donors <50 years old) are similar to those from standard deceased donors.
Age	Reduced allograft survival with extremes of donor age. Kidneys from older donors have fewer functioning nephrons due to aging and conditions such as hypertension and atherosclerosis. Younger donors (<5 years) are associated with worse outcomes, reflecting higher rates of technical complications.
Sex	Kidneys from female donors have slightly poorer survival probably due to nephron underdosing due to smaller kidney mass in women than men.
Race/Ethnicity	Deceased donor kidneys from AA are associated with reduced allograft survival largely relating to the presence of the APOL1 high-risk variant (in ~15% of AA donors). AA kidneys from donors lacking the APOL1 high-risk variant are comparable to non-AA donated kidneys. Integration of APOL1 genotyping (instead of race) into clinical prediction models (e.g., KDPI) is warranted.
Recipient Factors	
Timing	Preemptive transplantation is associated with less acute rejection and allograft failure. Longer time on dialysis is independently associated with poorer patient and graft survival.
Age	Graft survival rates are poorer in those at the extremes of age (<18 years; >65 years). In the very young, technical causes of graft loss (e.g., vessel thrombosis) and acute rejection are more common; death with a functioning graft is rare. In elderly recipients, death with a functioning graft is much more common (~ >50% of graft failures); acute rejection is less common.
Sex	Female recipients experience more graft loss and higher excess mortality than male recipients, particularly when the donor is male. Females are at higher rejection risk and have higher immune sensitization than males (primarily due to pregnancy).
Race/Ethnicity	AA recipients have worse graft survival than white recipients reflecting multiple factors including more DGF, more acute and late rejection, stronger immune responsiveness, a predominantly white donor pool (with resultant poorer matching of HLA and non-HLA antigens), altered pharmacokinetics of immunosuppressive drugs, differences in socioeconomic factors, and a higher prevalence of hypertension. Asian and Latino recipients have superior outcomes to white recipients; it is unknown why.
PRA	Patients who are highly sensitized are at higher risk for adverse early and late graft outcomes compared with unsensitized recipients.
DGF	DGF is associated with poorer patient and allograft survival. Allograft half-life is reduced by 30% with DGF, a larger effect than early acute rejection.
Compliance	Poor compliance with the immunosuppressive regimen markedly increases the risk of acute rejection (particularly late acute rejection) and allograft loss. Allograft loss has been reported to be 7-fold higher in nonadherent patients.
Donor-Recipient Factors	
HLA MM	Greater degree of HLA mismatch increases rejection risk and reduces allograft survival. HLA epitope-matching may prove important in coming years, though not widely used at present.
CMV matching	CMV D–/R– pairings have the best outcomes, D+/R– pairings have the worst. CMV probably affects graft outcomes through overt infection, but subclinical effects on immune function may also be important.
Sex matching	Female recipients of male kidneys have worse graft survival relating to an immune response to antigens on the Y-chromosome (H-Y antigens). The generally larger nephron dose in male donors may be a confounding factor in registry analyses.
Size matching	An imbalance between the metabolic demands of the recipient and the functional transplant mass (nephron load) may contribute to development and progression of chronic allograft injury when the recipient size is larger than the donor (>30 kg)
Surgical Factors	
Center effect	Outcomes vary from transplant center to center reflecting normal statistical variance, as well as center expertise and culture.
CIT	Prolonged CIT is associated with higher risk of DGF and poorer allograft survival.
WIT	Prolonged WIT is associated with higher risk of DGF and poorer allograft survival.

AA, African Americans; *CIT,* cold ischemic time; *CMV,* cytomegalovirus; *D,* donor; *DCD,* donation after cardiac death; *DGF,* delayed graft function; *HLA,* human leukocyte antigen; *KDPI,* kidney donor profile index; *NDD,* neurologic determination of death; *PRA,* panel reactive antibody; *R,* recipient; *WIT,* warm ischemic time.

Table 69.8 Cardiovascular Risk Factor Management in Kidney Transplant Recipients[175,300,305,416-437]

Risk Factor	Goal of Treatment	Treatment Recommendation	Comments
Posttransplant diabetes	Target HbA1c 7.0-7.5%; avoid HbA1c ≤6.0% especially if hypoglycemia reactions are common	Lifestyle modification; oral agents; insulin	Avoid metformin with reduced eGFR due to risk of lactic acidosis; Emerging data on SGLT-2 inhibitor use in transplant recipients are pending
Hypertension	<130/<80 mm Hg	Lifestyle modification; calcium channel blockers are first line; ACEi/ARB if proteinuria, diabetes of CVD; β-blocker if CVD	Nondihydropyridine CCB (verapamil and diltiazem) interact with CNIs; elevated CNI levels often require dose adjustment
Proteinuria	Reduce proteinuria	Lifestyle modification; when urine protein excretion ≥1 g/day for ≥18 years old and ≥600 mg/m^2/24 h consider ACEi/ARB as first line	Clinically important reductions in proteinuria with RAAS blockade, but no clear evidence of reduced all-cause mortality, transplant failure, or doubling of serum creatinine
Dyslipidemia	Adults: total cholesterol <200; LDL <100; non-HDL <130; Triglycerides <150 Adolescents: LDL<130; Non-HDL <160	Lifestyle modification; statins; fibrates; ezetimibe Patients with a kidney transplant should receive statin therapy, irrespective of lipid profile	Starting dose of a statin should be reduced as CNIs (especially cyclosporine) increase statin blood—increased risk of statin-related toxicities
Obesity	BMI <35-40 kg/m^2	Lifestyle counseling including an assessment of psychosocial stressors, sleep hygiene Increased physical activity and caloric restriction; bariatric surgery	Increased complications with bariatric surgery post transplant indicate more research needed; emerging practice to use GLP-1 RA (data on risk/benefit in transplant recipients are pending)
Tobacco use	Smoking cessation	Psychologic and/or pharmacologic support	Risk of weight gain, lifestyle counselling as above.

ACEi, Angiotensin-converting enzyme inhibitor; *ARB,* angiotensin receptor blocker; *BMI,* body mass index; *CVD,* cardiovascular disease; *GLP-1 RA,* glucagon-like peptide-1 receptor agonists; *HbA1c,* hemoglobin A1c; *HDL,* high-density lipoprotein; *LDL,* low-density lipoprotein; *SGLT-2,* sodium-glucose cotransporter 2.

From Kidney Disease: Improving Global Outcomes (KDIGO) Transplant Work Group. KDIGO clinical practice guideline for the care of kidney transplant recipients. *Am J Transplant*. 2009;9(Suppl 3):S1–S155.

5 mL/min/1.73 m^2 below an eGFR of 45 mL/min/1.73 m^2).[325] Transplant recipients experience more frequent fatal arrhythmias than the general population,[239] and traditional risk prediction scores (Framingham Risk Score) underestimate cardiovascular risk in kidney transplant recipients.[326] Furthermore, kidney transplant recipients are often systematically excluded from CVD trials, and the quality of evidence guiding CVD preventative or treatment strategies in this population is poor, with most recommendations extrapolated from nontransplant populations.[314] This may explain the poor uptake of preventative strategies in kidney transplant recipients.[327] Despite the limited evidence guiding recommendations, suggestions for management of cardiovascular risk in kidney transplant recipients are shown in Table 69.8.

POSTTRANSPLANT DIABETES MELLITUS

Posttransplant diabetes mellitus (PTDM) has a 3-year cumulative incidence of >40%, with the highest risk occurring in the first 6 months after transplant (12%–21%).[328,329] Risk factors include older age, obesity, positive hepatitis C antibody, CMV infection status, non-White ethnicity, family history, steroids, CNIs (especially tacrolimus), and episodes of acute rejection. Strategies to prevent and treat PTDM include steroid minimization, avoidance of tacrolimus, and lifestyle modification.[330,331] PTDM is associated with an increased risk of cardiovascular disease, infection, health care expenditure, and reduced patient and graft survival.[328,332,333] PTDM may require treatment with oral agents or insulin. Metformin is the drug of choice in kidney transplant recipients with adequate GFR[334]; however, no long-term prospective trials have examined the optimal treatment for PTDM, and given the lack of guiding evidence,[314] the 2020 Kidney Disease Improving Global Outcomes (KDIGO) guidelines do not specify glycemic targets, therapies, or monitoring recommendation.[335] Similarly, guidelines from the United Kingdom regarding management of PTDM are largely based on evidence graded as low quality and extrapolated from findings in the general population. For nontransplant patients with type 2 diabetes mellitus, the American Diabetes Association recommends an HbA1C target of 7%. There are certainly benefits of tight glycemic control, especially in terms of the reduction in microvascular complications.[336,337] However, these benefits must be weighed against the risks of overzealous glycemic control as evidenced by the ACCORD trial, where intensive glycemic control (targeting a HbA1C <6%) was associated with an increased risk of mortality and severe hypoglycemia.[336,338]

HYPERTENSION

The prevalence of hypertension amongst kidney transplant recipients in the CNI era is at least 60% to 90%.[339,340] Causes include use of steroids, CNIs, weight gain, allograft dysfunction, native kidney disease, and transplant renal artery stenosis. The complications of posttransplant hypertension are presumed to be a heightened risk of cardiovascular disease and allograft failure.[341] A post hoc analysis of the FAVORIT trial (which investigated the effect of lowering homocysteine levels on cardiovascular outcomes in kidney transplant recipients) found that each increase of 20 mm Hg in baseline systolic blood pressure was associated with 32% and 13% increases in the adjusted relative risks of cardiovascular events and death, respectively. In contrast, each 10 mm Hg drop in diastolic blood pressure below 70 mm Hg was associated with a 31% increase in the relative risks of both cardiovascular events and death. The highest risk was seen in patients with a systolic blood pressure >140 mm Hg and diastolic blood pressure <70 mm Hg.[342]

There has been some controversy over blood pressure treatment targets in recent years with no long-term, prospective trials to examine optimal blood pressure targets for kidney transplant recipients. 2021 KDIGO guidelines therefore do not include definitive recommendations for blood pressure targets in kidney transplant recipients, but rather a practice point suggests targeting a blood pressure of <130/<80 mm Hg.[343]

Nonpharmacologic measures such as weight loss, reduced sodium intake, reduced alcohol intake, treatment of obstructive sleep apnea, and increased exercise should be encouraged. However, antihypertensive drug therapy is still frequently required. KDIGO recommends first-line antihypertensive therapy be with a dihydropyridine calcium channel blocker or an angiotensin receptor blocker for kidney transplant recipients (grade 1C),[343] both of which have been shown to improve graft (but not necessarily patient) survival in randomized trials. Importantly, nondihydropyridine CCB (verapamil and diltiazem) interact with CNIs via inhibition of CYP 3A and intestinal P-gp activity, resulting in elevated CNI levels often requiring dose adjustment.[344] Although the evidence supporting RAAS blockade is controversial with no clear evidence of improved patient or graft survival versus control,[345] after the first few months post transplant, ACEIs or ARBs are routinely used in patients with proteinuria, diabetes, or other cardiovascular indications. Factors that influence choice of antihypertensive therapy in kidney transplant recipients are shown in Table 69.9.

PROTEINURIA

Proteinuria, even when modest, is associated with reduced allograft survival and an increase in both cardiovascular and noncardiovascular mortality.[346] In the general population, RAAS blockade reduces proteinuria, hypertension, progression to ESKD, and mortality. However, despite clinically important reductions in blood pressure and proteinuria,[347] systematic reviews in kidney transplantation have failed to demonstrate evidence of reduced all-cause mortality, transplant failure, or doubling of serum creatinine with RAAS inhibition compared with placebo.[345] Notably, trials of RAAS blockade in transplant populations have been limited by small numbers and event rates, and relatively short follow-up and results must be interpreted in this context; we currently initiate ACEI or ARB treatment in kidney transplant recipients with proteinuria. Given the vast benefits with SGLT-2 inhibitors in the general population (especially in those with proteinuria), these agents may show eventual promise in treating proteinuric transplant recipients, but to date data are insufficient regarding their risk/benefit to recommend use in this unique population.

HYPERLIPIDEMIA

The prevalence of hyperlipidemia after transplant is high.[348] Steroids, CNIs (cyclosporine > tacrolimus), and sirolimus are considered principal causes or contributors. Dyslipidemia has been associated with increased cardiovascular mortality, graft loss, and allograft rejection in patients with a kidney transplant.[349] While statin use effectively improves the lipid profile in kidney transplant recipients, this has not consistently translated into reduced CVD mortality in the transplant population.[350] In the ALERT trial, the use of fluvastatin versus placebo in kidney transplant recipients did not significantly reduce the composite of cardiac death, nonfatal MI, or coronary intervention procedure (primary endpoint). However, it did improve isolated cardiac mortality as a secondary outcome,[351] and a follow-up study after an additional 2 years of fluvastatin therapy showed a significant reduction in a composite of cardiac death and nonfatal MI.[352] Hypertriglyceridemia is also a challenge post transplant. Strategies for lowering triglyceride levels include lifestyle modification, substitution of sirolimus for an alternative agent (e.g., MMF), and treatment with ezetimibe. In the largest kidney transplant RCT to date (FAVORIT), high-dose folic acid/B6/B12 versus placebo in a population of stable kidney transplant recipients with elevated homocysteine levels showed a reduction in homocysteine but not cardiovascular disease, mortality, or dialysis-dependent kidney failure.[353] KDIGO recommends that patients with a kidney transplant receive statin therapy, irrespective of their lipid profile (grade 2B evidence).[354] The application of a risk-guided statin treatment algorithm in the kidney transplant population seems like a reasonable approach. The only caveat is that the starting dose of a statin should be reduced in kidney transplant recipients as the CNIs (especially cyclosporine) increase statin blood levels and may predispose to statin-related toxicities.[355]

OBESITY

The prevalence of obesity among kidney transplant recipients (and the general population) is increasing over time; the number of people classified as obese doubles over the first year post transplant.[356] This may relate in part to steroid therapy, which is a mainstay of many maintenance immunosuppressive regimens, although many studies suggest weight gain after transplant also occurs independent of steroid therapy and dose. Liberalization of the restrictive “kidney diet” and improved appetite with increased caloric intake may play a role. Obesity in transplant recipients has been associated with a greater risk of perioperative and postoperative complications, PTDM, hypertension, and reduced allograft and patient survival.[357] Nevertheless, studies of patients with BMI >30 kg/m^2 suggest that transplantation provides a survival benefit over remaining

Table 69.9 Antihypertensive Agents in the Transplant Recipient

Indication	ß-blocker	ACE-I/ARB	Ca-Channel Blocker	Diuretic Thiazide
Hypertension only		X	X	X
CHF	X	X		
Post-MI	X	X		
CAD	X	X		
Diabetes		X		
Proteinuria		X		

ACE-I, Angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; CAD, coronary artery disease; CHF, congestive heart failure; MI, myocardial infarction.

Table 69.10 Cancers Categorized by Standard Incidence Rate (SIR) for Kidney Transplant Patients and Cancer Incidence

	Common Cancers in Both General and Transplant Populations: Incidence >10 Per 100,000 People	Common Cancers in Transplant Population: Incidence in General Population <10 Per 100,000, but Estimated Incidence in Transplant Population >10 Per 100,000 People	Rare Cancers Incidence in Both General and Transplant Populations <10 Per 100,000 People
High SIR >5	Kaposi sarcoma (with HIV)	Kaposi sarcoma Vagina Non-Hodgkin lymphoma Kidney Nonmelanoma skin Lip Thyroid Penis Small intestine	Eye
Moderate SIR (1-5)	Lung Colon Cervix Stomach Liver	Oronasopharynx Esophagus Bladder Leukemia	Melanoma Larynx Multiple myeloma Anus Hodgkin lymphoma
No Increased Risk	Breast Prostate Rectum		Ovary Uterus Pancreas Brain Testis

Adapted from the KDIGO practice gudelines.

on the waiting list (on dialysis).[6] Interestingly, posttransplant weight gain appears to be associated with greater risk to the patient and allograft than obesity present in candidates at the time of transplant.

In the general population, lifestyle counseling including an assessment of psychosocial stressors, sleep hygiene, and other individualized approaches has been recommended, in addition to increased physical activity and caloric restriction.[358] However, few studies have investigated weight loss strategies in kidney transplant recipients with obesity. KDIGO has recommended offering conservative weight-reduction programs to all kidney transplant recipients living with obesity. For patients with refractory obesity after transplant, bariatric surgery has been associated with sustained weight loss and improvement in graft function, but the limited studies in kidney transplant recipients have demonstrated a potential signal for increased complications and mortality that indicates more research is needed.

SMOKING

Tobacco smoking should be strongly discouraged; there is evidence that it affects allograft function, as well as recipient survival.[359]

CANCER AFTER KIDNEY TRANSPLANTATION

Kidney transplant recipients are at a 2 to 4× higher risk of cancer and cancer-related mortality than the general population (Table 69.10),[360-362] with a >25% risk of experiencing a solid organ cancer at 20 years post transplant.[363] A review of outcomes in Australia and New Zealand has shown that among transplant recipients, the relative reduction in cardiovascular mortality over time

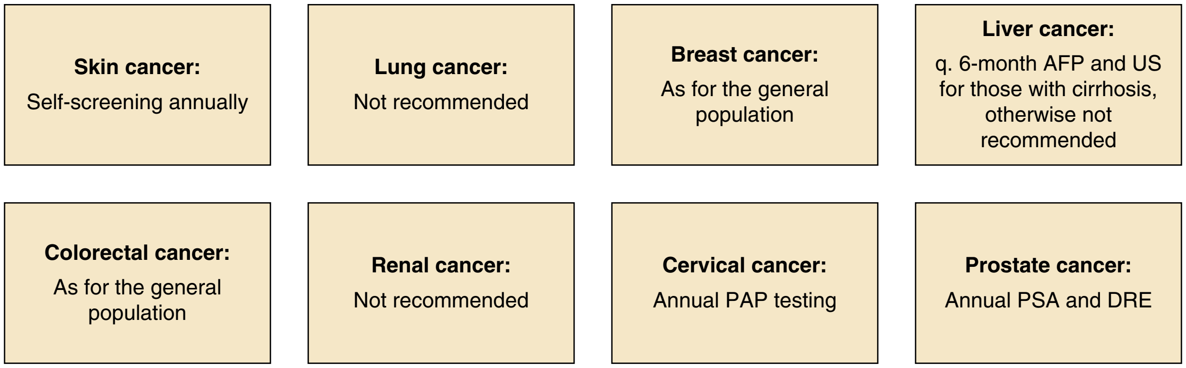

Fig. 69.6 Cancer screening recommendations for kidney transplant recipients. *PAP,* Papanicolaou test; *PSA,* prostate specific antigen; *DRE,* digital rectal exam. *Based on the most common recommendations from the AST (2000), KDIGO (2009), EBPG (2002), CST (2010), NKF (2009) and RA (2011) guidelines.*

has resulted in a relative (but not absolute) increased rate of late (>1 year) cancer deaths.[315] For many cancer types, transplant recipients present with more advanced-stage disease; however, survival is lower in transplant patients at any cancer stage than in the age- and sex-matched general population. Cancer screening recommendations for kidney transplant recipients are shown in Fig. 69.6. Interestingly, unlike other cancer types, breast and prostate cancer risk does not appear to be increased in transplant recipients.

There are several reasons why transplant recipients are at increased risk for de novo and recurrent malignancy, including an increased susceptibility to oncogenic viruses (Table 69.11), immunosuppression, and altered T cell immunity.[363] Immunosuppression inhibits normal tumor surveillance mechanisms, allowing unchecked proliferation of "spontaneously occurring" neoplastic cells. Risk factors for malignancy after transplant include advanced age, smoking, male sex, White ethnicity, and prolonged dialysis vintage before transplant.[363] It is believed that cumulative immunosuppression exposure rather than a specific drug is the most important factor increasing cancer risk after transplant, though notably T cell depleting agents have been shown to increase the risk of non-Hodgkin lymphoma by 30% to 80% and anti-CD25 monoclonal antibody induction (alemtuzumab) is associated with a 70% to 200% higher incidence of malignancy.[363] The single most important measure to prevent cancers is to minimize excess immunosuppression. When cancer occurs, immunosuppression is generally decreased. In some cases, rejection of the allograft may result; the risks and benefits of immunosuppression must be judged on a case-by-case basis.

SKIN AND LIP CANCERS

Squamous cell carcinoma (SCC), basal cell carcinoma, and malignant melanoma are all more common in kidney transplant patients. The incidence of nonmelanomatous skin and lip cancer in kidney transplant recipients is ~2 to 40× greater than in the general population,[361,363] and the routine use of CNIs has increased this risk over time.[364] Azathioprine may further increase DNA sensitization to ultraviolet A radiation and has been independently associated with a twofold to ninefold increase in the risk of squamous cell carcinoma.[365] Nonmelanomatous skin cancers cause immense morbidity but are rarely fatal.

Table 69.11 Viral Infections Associated With Development of Cancers in Kidney Transplant Patients

Virus	Neoplasm
HBV	Hepatocellular cancer
HCV	Hepatocellular cancer
EBV	PTLD
HPV	Squamous cell cancers of anogenital area and of mouth
HHV-8	Kaposi sarcoma

EBV, Epstein-Barr virus; *HBV,* hepatitis B virus; *HCV,* hepatitis C virus; *HHV-8,* human herpes virus-8; *HPV,* human papillomavirus; *PTLD,* posttransplant lymphoproliferative disorder.

Risk factors include longer time since transplant, cumulative immunosuppressive dose, exposure to ultraviolet light, fair skin, and human papillomavirus (HPV) infection. Primary and secondary prevention is important: Patients should be specifically counseled on minimizing lifelong exposure to ultraviolet light (e.g., avoidance of sunlight exposure during peak hours, wearing protective clothing including wide-brimmed hats, the use of UV blocking sunscreen) and to self-screen for skin lesions.[339] KDIGO additionally recommends annual skin and lip examination by dermatologists or other qualified health care professionals (grade 2D evidence).[339] Retinoids are sometimes used in high-risk patients.[366] Suspicious skin lesions should be surgically excised. In patients with posttransplant SCC, switching from a CNI to sirolimus reduces the risk of further SCC by 44%.[137]

ANOGENITAL CANCERS

Cancers of the anogenital region are significantly more common in kidney transplant recipients (standard incidence ratio >5). These cancers tend to be multifocal and more aggressive than in the general population. Infection with certain HPV strains (e.g., 16 and 18) is an important

Table 69.12 Clinical and Pathologic Spectrum of Posttransplant Lymphoproliferative Disorder (PTLD) and Summary of its Management

	Early Disease (50%)	Polymorphic PTLD (30%)	Monoclonal PTLD (20%)
Clinical features/ symptoms	Infectious mononucleosis-type illness	Infectious mononucleosis-type illness ± weight loss, localizing symptoms	Fever, weight loss, localizing symptoms
Pathology	Preserved architecture, atypical cells infrequent	Intermediate transformed cells	High-grade lymphoma with confluent and marked atypia
Clonality	Polyclonal	Usually polyclonal	Monoclonal
Treatment	Reduce immunosuppression	Reduce immunosuppression ±rituximab	Reduce immunosuppression ± rituximab ± chemotherapy
Prognosis	Good	Intermediate	Intermediate/poor

PTLD, Posttransplant lymphoproliferative disorder.

risk factor. Secondary prevention measures include yearly physical examination of the anogenital area and, in women, yearly pelvic examinations and cervical histology. Suspicious lesions should be excised, and patients should be closely followed for recurrence. HPV vaccination has reduced the incidence of anogenital and oropharyngeal cancers in the general population, and although safety and efficacy data are limited after transplantation, it is also expected to benefit transplant recipients, particularly when they are vaccinated before transplant.[367]

KAPOSI SARCOMA

While rare overall, Kaposi sarcoma exhibits an incidence as high as 300-fold greater in transplant patients than the general population,[368] with a risk that is strongly associated with ethnic background. Those of Jewish, Arab, and Mediterranean ancestry are at much higher risk. Other risk factors are cumulative immunosuppressive dose and human herpesvirus-8 infection. Visceral (lymph nodes, lungs, gastrointestinal tract) and nonvisceral (skin, conjunctivae, oropharynx) involvement may occur. The prognosis for the former is poor, but for the latter it is good. Treatment involves various combinations of surgical excision, radiotherapy, chemotherapy, and immunotherapy with immunosuppression reduced or modified. Substitution of sirolimus for CNIs has been reported to slow progression of the disease without sacrificing allograft function.[135]

POSTTRANSPLANT LYMPHOPROLIFERATIVE DISORDER

PTLD can occur early after transplant and carries a high morbidity and mortality. The cumulative 3-year incidence of PTLD is 0.5% in adult and 1.5% in pediatric kidney transplant recipients, a rate 4- to 16-fold higher in transplant recipients than in the general population.[363]

PTLD involves a range of abnormal lymphoproliferations from benign polyclonal lymphoid proliferation to aggressive malignant monoclonal lymphomas. Epstein-Barr virus (EBV) is important in the pathogenesis of PTLD including the infection and transformation of B cells by EBV in the setting of immunosuppression and ineffective T cell surveillance. Transformed B cells undergo proliferation that is initially polyclonal, but a malignant clone may evolve. Thus the clinical and histologic spectrum of PTLD at presentation and its treatment can vary greatly (Table 69.12). Despite the strong association with EBV, the proportion of EBV-negative PTLD cases has increased over time.[369]

Risk factors for PTLD include 1. EBV-positive donor and EBV-negative recipient; 2. CMV-positive donor and CMV-negative recipient; 3. pediatric recipient (in part because children are more likely to be EBV naive); and 4. intensity of immunosuppression.[370] Clinical trials of belatacept saw high rates of PTLD in EBV-negative subjects, as discussed earlier; while controversial, the drug is considered by many as contraindicated in patients who are seronegative for EBV.[107] Extranodal, gastrointestinal tract, and central nervous system involvement is more common than in nontransplant lymphomas, and the kidney allograft may be involved. More than 90% are non-Hodgkin lymphomas, and most are of recipient B cell origin.[371] Because early primary EBV infection post transplant is a strong risk factor for PTLD, screening high-risk patients (EBV-positive donor/EBV-negative recipient) for EBV viremia in the first year post transplant is recommended. EBV viremia should prompt reduction of immunosuppression, which is believed to prevent and treat PTLD occurrences. The diagnostic work-up for PTLD includes imaging (CT and PET scans); histopathologic diagnosis requires biopsy of bone marrow, concerning masses, or lymph nodes.[372] Treatment options for established PTLD include immunosuppression reduction, rituximab, and CHOP (cyclophosphamide, doxorubicin [Adriamycin], vincristine, prednisone).

INFECTIOUS COMPLICATIONS OF KIDNEY TRANSPLANTATION

The transplant procedure and subsequent immunosuppression increase the risk of serious infection. The principal factors determining the type and severity of infection are the degree of immunosuppression and exposure (in the hospital and community) to potential pathogens.[373] Factors affecting the net state of immunosuppression include the cumulative amount of immunosuppression, recipient comorbidities (e.g., diabetes, UTIs), infection with viruses that affect the immune system (e.g., EBV, CMV, HIV, hepatitis C), and the integrity of mucocutaneous barriers.

The patterns of infection after kidney transplantation can be roughly divided into three time periods: 0 to 1 month, 1 to 6 months, and more than 6 months after transplant.[373] These divisions of time serve as guidelines only. An outline

<1 month

Infection with antimicrobial-resistant species:
MRSA, VRE, CRE, candida species (non-albicans)
Aspiration
Catheter infection
Wound infection
Anastamotic leaks and ischemia
Clostridium difficile colitis
Donor-derived infection (uncommon):
HSV, LCMV, rhabdovirus (rabies), West Nile virus, HIV, T. *cruzi*
Recipient-derived infection (colonization):
Aspergillus, pseudomonas

1–6 months

With PJP and antiviral (CMV, HBV) prophylaxis:
Polyomavirus, BK infection, nephropathy
Clostridium difficile colitis
HCV infection
Adenovirus infection, influenza
Cryptococcus neoformans infection
Mycobacterium tuberculosis infection
Without prophylaxis:
Pneumocystis
Infection with herpesviruses [HSV, VZV, CMV, EBVI
HBV infection
Infection with *Listeria, Nocardia,* toxoplasma, strongyloides, leishmania, T. *cruzi*

>6 months

Community-acquired pneumonia, UTI
Infection with aspergillus, typical molds, mucor species
Infection with nocardia, rhodococcus species
Late viral infections:
CMV infection
Hepatitis (HBV, HCV)
HSV encephalitis
Community-acquired (SARS, West Nile virus infection)
JC polyomavirus infection (PML)
Skin cancer, lymphoma (PTLD)

Fig. 69.7 Pattern of infection after transplantation according to time after transplantation.

of the pattern of infection after transplantation according to time after transplantation is shown in Fig. 69.7.

Importantly, infection in transplant recipients may progress rapidly and present atypically; signs and symptoms of infection may be diminished, and it can be difficult to recognize even serious infectious processes in the setting of immunosuppression.[373] Conversely, noninfectious conditions including acute allograft rejection may present with fever and systemic inflammatory-type features, making the diagnosis often difficult. Prompt diagnosis often includes invasive diagnostic procedures (e.g., bronchoscopy in patients with pneumonitis), and targeted antimicrobial therapy is essential.[373] Many antimicrobial agents interact with immunosuppressive therapies, and sepsis itself can induce changes in immunosuppressive pharmacokinetics and pharmacodynamics; therefore lability of drug levels and potential toxic interactions must be closely monitored for. There is no consensus regarding the management of immunosuppressive agents during severe infection, though abrupt cessation or rapid withdrawal of glucocorticoids should be avoided given the potential of precipitating adrenal insufficiency; stress-dose steroids should be considered in the setting of septic shock.[374] Judicious withdrawal of immunosuppression may be indicated for certain viral infections or in the setting of difficult to control bacterial or fungal infections; however, the risk of allograft rejection must be considered. Furthermore, emerging evidence suggests immunosuppression may have some beneficial effects in protecting septic patients from major downstream inflammatory and coagulation responses to severe infections, which have been associated with subsequent mortality.[375] The relative risk-benefit of immunosuppression reduction in these situations remains unclear.

INFECTIONS IN THE FIRST MONTH

The majority of infections in the first month are standard infections as would be seen in nontransplant patients after surgery. Thus infections of surgical wounds, the lungs and urinary tract, and infections related to vascular catheters predominate. Bacterial infections are much more common than fungal ones. Early vial reactivation (e.g., HSV) is possible, but the risk is reduced by posttransplant viral prophylaxis. Given modern screening strategies, donor-derived infections are rare but typically present in the first month after transplant. Preventive measures include ensuring that donor and recipient are free of overt infection before transplant, good surgical technique, and SMX-TMP prophylaxis to prevent UTIs.

INFECTIONS FROM 1 TO 6 MONTHS AFTER TRANSPLANT

The intermediate period posttransplant is felt to be the greatest risk for opportunistic infections. Given the need for relatively high immunosuppression requirements during months 1 to 6 post transplant, the risk of bacterial, viral, fungal, and protozoal infections is increased. Infections with CMV, EBV, *Listeria monocytogenes, Pneumocystis jirovecii,* and *Nocardia* spp. are relatively common. Preventive measures include antiviral prophylaxis (for 3–6 months) and SMX-TMP or PJP prophylaxis (for 6–12 months), discussed below.[373]

INFECTIONS MORE THAN 6 MONTHS AFTER TRANSPLANT

With gradual reduction in immunosuppression, the risk of infection long term usually diminishes. When infection does occur, it is typically due to a community-acquired bacterial or viral pathogen. Patients with stable ongoing allograft function and no need for late supplemental immunosuppression are generally at low risk for developing opportunistic infections unless exposure is intense (e.g., to *Nocardia* spp. from soil). In contrast, those with poor allograft function are at higher risk for opportunistic infection, probably reflecting both poor allograft function and the possible need for larger cumulative doses of immunosuppression. Late amplification

Table 69.13 Manifestations of Cytomegalovirus Disease in the Kidney Transplant Recipient

Tissue Affected	Clinical/Pathologic Features	Comment
Viremia (blood)	Fever, malaise, myalgia	Nonspecific, may occur after cessation of prophylaxis
Bone marrow	Leukopenia	May require azathioprine or MMF dose reduction. Valganciclovir may also cause leukopenia
Lungs	Pneumonitis	May be life-threatening; exclude coinfection with other organisms
GI tract	Inflammation and ulceration of esophagus or colon	May be life-threatening
Liver	Hepatitis	Rarely severe
Retina	Blurred vision, flashes, floaters	Relatively rare in kidney transplants; if occurs, usually late.
Kidney	Viral inclusions	Association with FSGS

GI, Gastrointestinal; MMF, mycophenolate mofetil.

of immunosuppression may increase the risk of opportunistic infection in any patient. Therefore any patient receiving a "late" steroid pulse or antithymoglobulin therapy should be restarted on SMX-TMP ± anti-CMV prophylaxis (if donor or recipient were CMV positive).[373] The implications of BKV and EBV infection have been discussed earlier; CMV, *P. jirovecii* pneumonia, and COVID infection are discussed below.

CYTOMEGALOVIRUS

Exposure to CMV (as evidenced by the presence in serum of anti-CMV IgG) increases with age; more than two-thirds of adult donors and recipients are latently infected before kidney transplantation. Infection may arise from 1. reactivation of latent recipient virus, 2. primary infection with donor-derived virus, or 3. reactivation of latent donor-derived virus. CMV *disease* means that there is infection with symptoms, with evidence of tissue invasion, or both. The risk of CMV infection or disease is highest in CMV-positive donor/CMV-negative recipient pairings, followed by CMV-positive donor/CMV-positive recipient pairings. The risk is lowest when both recipient and donor are negative for CMV at the time of transplant. Lymphocyte-depleting therapy for induction or when prescribed for treatment of rejection significantly increases the risk of subsequent CMV disease.

CMV disease usually arises 1 to 6 months after transplantation, although gastrointestinal and retinal involvement often occurs later. Viral prophylaxis may delay CMV infection until it has been discontinued, with postprophylaxis CMV infection typically occurring in months 6 to 12. CMV most commonly presents as asymptomatic viremia; however, clinical features may include fever, malaise, gastrointestinal disturbance, and leukopenia; there may be symptoms of, or laboratory evidence of, specific organ involvement (Table 69.13). Urgent investigation and immediate empiric treatment are needed in severe cases. Confirmation of presumed CMV disease is by demonstration of the virus in body fluids or solid organs. Quantitative CMV PCR is fast and sensitive and allows for monitoring of viral load in response to therapy. However, viremia may be absent in certain CMV infections, most notably involving the retina or gastrointestinal tract. In such instances, the CMV infection may be diagnosed by ophthalmologic examination ± vitreous humor CMV PCR and immunohistochemistry or culture of gastrointestinal tissue, respectively. Other procedures such as lumbar puncture and bronchoscopy should be aggressively pursued according to symptoms and signs. A "tissue diagnosis" is also required to exclude coinfection with other microbes such as *P. jirovecii*. In addition to direct effects, CMV may have indirect effects including an increased risk of coinfection, rejection, and PTLD.[376]

The treatment of posttransplant CMV infection depends on disease severity and the organ systems involved. Low-grade CMV viremia usually responds to a reduction in immunosuppression (usually the antimetabolite). The VICTOR trial demonstrated that oral valganciclovir was not inferior to intravenous ganciclovir in the treatment of non–life-threatening CMV infection in solid organ transplants.[377] Therefore patients with symptomatic or significant viremia but relatively mild symptoms may be treated as outpatients with oral valganciclovir. However, patients with severe systemic CMV infection or organ-specific disease (i.e., pneumonitis, gastrointestinal, or central nervous system [CNS]) should be treated initially with intravenous ganciclovir with a transition to oral valganciclovir as symptoms improve. CMV viral load should be monitored weekly with therapy, acknowledging that viral loads may not accurately reflect CMV disease severity. For viremic patients with mild to moderate disease, some centers continue maintenance treatment for 2 weeks after resolution of viremia, but the use and duration of maintenance therapy is controversial. Treatment of severe or invasive disease may benefit from longer courses of induction treatment followed by maintenance dose therapy.[378] Dose adjustment is required in kidney dysfunction for both valganciclovir and ganciclovir.

The prevention of CMV disease is of great clinical importance. One strategy is to give prophylaxis to all patients at risk (D+/R–, D–/R+, D+/R+). Another preemptive strategy is to monitor for CMV viremia and begin therapy only when there is evidence of active viral replication.[376] Prophylactic and preemptive strategies using valganciclovir have been shown to be comparable in terms of cost and prevention of symptomatic CMV.[379] Valganciclovir has proven to be an excellent therapy for both the prophylaxis and treatment of CMV; however, myelosuppression, especially leukopenia/neutropenia, is a common and sometimes difficult-to-manage side effect. Letermovir is a newer agent that appears to be as effective as valganciclovir in preventing CMV infection post transplant, with fewer cytopenias.[380] Maribivir is emerging as a potentially useful agent in the treatment of refractory or resistant CMV, with less toxicity than existing agents.[381]

PNEUMOCYSTOSIS

Antimicrobial prophylaxis is effective in preventing pneumonia due to *P. jirovecii* (previously *carinii*); before the routine use of prophylaxis, the incidence of *P. jirovecii* post transplant was ~10% to 15%. The preventive agent of choice is SMX-TMP. It is generally well tolerated and quite inexpensive; furthermore, it prevents UTIs and opportunistic infections such as nocardiosis, toxoplasmosis, and listeriosis. Alternative preventive agents include dapsone and pyrimethamine, atovaquone, and aerosolized pentamadine.[382] Prophylaxis is recommended for 6 to 12 months after transplant and after escalation of immunosuppression for the treatment of acute rejection.[383] Risk factors include lymphopenia, CMV infection, hypogammaglobulinemia, older age, and treatment for acute rejection.[383] Typical symptoms of pneumonia due to *P. jirovecii* are fever, shortness of breath, and cough; however, *P. jirovecii* confers a significant risk to the patient and allograft, with a mortality as high as 50% even with appropriate therapy.[384] Chest radiography characteristically shows bilateral interstitial-alveolar infiltrates. Laboratory findings generally include a PO_2 <60 and elevated lactate dehydrogenase (usually >300 IU/mL), PAO_2-PaO_2 gradient, and serum (1,3) beta-D-glucan levels (with high sensitivity and low specificity).[383] Diagnosis requires detection of the organism in a clinical specimen by colorimetric or immunofluorescent stains. Because the organism burden is usually lower than in HIV-infected patients, the sensitivity of induced sputum or bronchoalveolar lavage specimens is lower in kidney transplant recipients; tissue should be quickly obtained if these tests are negative and the clinical suspicion remains high.

The treatment of choice remains SMX-TMP.[382] There is no firm evidence to support the use of adjunctive corticosteroids during the early treatment phase of pneumocystosis in kidney transplant recipients; however, given the poor outcomes in patients with hypoxemia (PAO_2 <70 mm Hg on room air) early steroid administration is controversial but generally recommended.[383]

COVID

SARS-CoV-2, the virus responsible for the COVID-19 pandemic, has resulted in substantial morbidity and mortality at a global level, with the highest risk observed in immunosuppressed patients including kidney transplant recipients. Early in the pandemic, the mortality rate among transplant recipients was ~15% to 20%, with an estimated 5000 excess COVID-19 related deaths in solid organ transplant recipients in the Unites States over the first 13 months (~1 in 75 transplant recipients).[385] Despite concerns regarding impaired cellular and humoral immune responses to SARS-CoV-2 vaccination in transplant recipients, vaccination significantly reduces mortality and prevents or reduces severity of other COVID-19 complications in solid organ transplant recipients, albeit to a lesser degree than that observed for the general population.[386] Therefore vaccination against SARS-CoV-2 is strongly recommended for all transplant recipients. Given the rapidity of evolving evidence regarding COVID-19 prevention and treatment in the general population and transplant patients alike, national guidelines are regularly updated and should be referenced for up-to-date public health recommendations as needed.

IMMUNIZATION IN KIDNEY TRANSPLANT RECIPIENTS

Important general rules concerning immunization in kidney transplant patients are the following: 1. Immunizations should be completed at least 4 weeks before transplantation; and 2) live vaccines are generally contraindicated after transplantation. We follow the *IDSA Clinical Practice Guideline for Vaccination of the Immunocompromised Host*.[387] We pay special attention to pretransplant vaccination in asplenic patients and those who will or may require eculizumab therapy. Whenever possible, vaccination should occur before transplantation given the waning response to vaccination after transplant. Vaccines that are recommended and contraindicated are summarized in Table 69.14. Household contacts of transplant recipients should receive yearly immunization against influenza and COVID-19 vaccination as indicated on the basis of evolving public health guidelines.

SPECIAL CONSIDERATIONS IN THE TRANSPLANT RECIPIENT

PREGNANCY IN THE KIDNEY TRANSPLANT RECIPIENT

Fertility is often improved after kidney transplantation.[388] Although the risks of pregnancy and childbirth are greater in kidney transplant recipients compared with the general population, it is generally considered safe for the mother, fetus, and kidney allograft if the following criteria are met before conception: good general health for more than 18 months before conception, stable allograft function with a plasma creatinine <2.0 mg/dL (preferably <1.5 mg/dL), minimal hypertension, minimal proteinuria, immunosuppression at maintenance doses, and no dilation of the pelvicalyceal system on recent imaging studies.[389] Pregnancy in women with kidney disease is discussed further in Chapter 58.

A systematic review and meta-analysis of 50 studies that included data on 4706 pregnancies in 3570 KTRs, between the years 2000 and 2010, give insight into the relative risks of pregnancy post kidney transplant. The rate of live births was actually higher than that of the general population. However, KTR pregnancies were more likely to end with cesarean sections, preterm births (35.6 vs 38.7 weeks), and lower birth weights (2.4 kg vs. 3.3 kg). Preeclampsia and gestational diabetes were both more common in KTRs than the general population. More than 50% of KTRs were hypertensive during pregnancy. Overall, pregnancy did not

Table 69.14 Recommended and Contraindicated Vaccines in Transplant Recipients

Recommended	Comments	Contraindicated[a]
Influenza	Annually	Live varicella vaccine[b]
PCV13	Once	BCG
PPSV23	Repeat every 5 years, up to 3 doses/lifetime including one dose in those aged 65 years and older	Smallpox
Tdap/Td	Td booster every 10 years	Intranasal influenza
9vHPV	Only for ≤ 26 years of age	Live Japanese B encephalitis
HepA		Oral polio
HepB	Give 3-dose series if anti-HBS<10 mIUL/mL	MMR
Shingrix	Adults 50+ years old; 2 doses 2-6 months apart	
COVID-19	As per evolving local guidelines	
Asplenia or complement inhibitor therapy		Live oral typhoid Ty21a
MenACWY	2-dose series with booster every 5 years	Yellow fever
MenB	2-dose series, no booster	
Hib	If not already given	

[a]Not an exhaustive list, all live vaccines are generally contraindicated.
[b]Awaiting data on the non-live "Shingrix" (HZ/su) zoster vaccine.
9vHPV, Human papillomavirus vaccine; *COVID-19, Sars-CoV-2 vaccine (undefined); HepA,* hepatitis A vaccine; *HepB,* hepatitis B vaccine; *Hib, Haemophilus influenzae* type b conjugate vaccine; *MenACWY,* meningococcal (quadrivalent) conjugate; *MenB,* serogroup B meningococcal vaccines; *MMR,* measles, mumps, and rubella vaccine; *PCV13,* Pneumococcal conjugate vaccine (13-valent); *PPSV23,* pneumococcal polysaccharide vaccine (23-valent); *Shingrix,* herpes zoster vaccine; *Td,* tetanus and reduced diphtheria toxoids; *Tdap,* tetanus and reduced diphtheria toxoid, and acellular pertussis vaccine.

appear to associate with an increased risk of acute rejection or subsequent graft loss.[390]

All pregnant kidney allograft recipients should be managed as high-risk obstetric cases with nephrology involvement. Throughout the pregnancy, regular monitoring of blood pressure, proteinuria, kidney function, and urine cultures is advised. Significant kidney dysfunction occurs in a minority of cases; the principal causes are severe preeclampsia, acute rejection, acute pyelonephritis, and recurrent glomerulonephritis. Distinguishing these causes clinically may be difficult. Initial investigations should include plasma creatinine, creatinine clearance, 24-hour urinary protein excretion, urine microscopy, urine culture, and kidney ultrasound. Acute rejection should be confirmed by allograft biopsy before instituting antirejection therapy. Pulse steroids are used to treat rejection.

There are no transplant-specific reasons to perform cesarean section; if it is performed (for obstetric reasons), care should be taken to avoid damaging the transplant ureter. Kidney function should be monitored closely for 3 months postpartum because of the increased risk of hemolytic uremic syndrome and possibly acute rejection.

Short-term and long-term data indicate that children born to transplant recipients using cyclosporine, steroids, or azathioprine do not have a significant increase in morbidity. Short-term data on tacrolimus are similarly reassuring. Dosages of cyclosporine and tacrolimus may need to be increased to maintain prepregnancy trough concentrations. CNI levels typically decrease during the first trimester due to increased volume of distribution. Therefore CNI levels should be closely monitored, and doses should be adjusted to avoid acute rejection from subtherapeutic levels. MMF is considered to be teratogenic, and it should not be used in women contemplating pregnancy.[391] Sirolimus should also be avoided.

SURGERY IN THE KIDNEY TRANSPLANT RECIPIENT

ALLOGRAFT NEPHRECTOMY

Allograft nephrectomy is not commonly required. Indications include: 1. allograft failure with ongoing symptomatic rejection causing fever, malaise, and graft pain; 2. infarction due to thrombosis; 3. severe infection of the allograft such as emphysematous pyelonephritis; and 4. allograft rupture. The morbidity associated with allograft nephrectomy is relatively high. Ongoing rejection in a failed allograft can sometimes be controlled with steroids, but prolonged immunosuppression of a patient who is on dialysis is obviously not ideal. Rejection in this context is less likely to be controlled by small doses of steroids than when it is acute and when the transplant is recent. A retrospective analysis of registry data noted a survival advantage in transplant recipients who underwent allograft nephrectomy after returning to dialysis.[392] However, any potential benefits of nephrectomy must be weighed against the heightened risk of HLA-sensitization around the time of allograft nephrectomy.[393] This may be of great importance for those patients listed for repeat kidney transplant. We generally only perform nephrectomy for symptomatic late rejection.

NONTRANSPLANT-RELATED SURGERY OR HOSPITALIZATION

Many kidney transplant recipients are hospitalized or undergo surgery for nontransplant reasons. Precautions for procedures and surgery in transplant recipients are listed in **eTable 69.4**. Common sense measures such as maintenance of adequate volume status, avoidance of nephrotoxic medicines (including nonsteroidal antiinflammatory drugs), and proper dosing of immunosuppressive drugs are advised. Whenever possible, immunosuppressive drugs should be given by the enteral route. When not possible, we use intravenous steroid and intravenous MMF or AZA.

eTable 69.4 Precautions for Procedures and Surgery in Kidney Transplant Recipients

- Caution with radiocontrast exposure
- Maintain hydration
- Avoid nephrotoxic antibiotics and analgesic
- "Stress-dose steroids" is not always necessary
- If enteral route of medication is contraindicated, give tacrolimus sublingually if able (1/2 total oral dose)
- If enteral route of medication is contraindicated/sublingual tacrolimus is not available, give calcineurin inhibitor via intravenous route (1/3 total oral dose for cyclosporine; 1/5 total oral dose for tacrolimus)
- Monitor allograft function, plasma potassium, and acid-base balance daily
- Consider wound-healing impairment with sirolimus

Sublingual tacrolimus may be used, usually at half the oral dose, instead of intravenous tacrolimus, but the instructions for administration must be adhered to, such as not swallowing for 5 to 15 minutes and avoiding oral intake for 15 to 30 minutes, which may be challenging in some patients. When necessary, intravenous tacrolimus should be started at one-fifth to one-third of the total daily dose, administered over 24 hours. Intravenous cyclosporine should be prescribed in slow infusion over 24 hours at one-third of the total daily oral dose.

TRANSPLANTATION/ IMMUNOSUPPRESSION: THE FUTURE

There are ongoing unmet needs in solid organ transplantation and in kidney transplant in particular. The gap between organ demand and supply is growing and has led to newer practices including updated organ allocation schema and expanded use of DCD organs. In the past decade, ex vivo perfusion (EVP) has moved from the experimental to the clinical realm to both ameliorate organ dysfunction from NDD and DCD donors and more aspirational goals to regenerate cell structure to take an unused organ into the quality that provides clinical function. EVP requires pump technology currently available for all organs including heart, lung, and liver. Under debate now are the temperature of the perfusate,[394] the contents of the perfusate,[395] cellular and genetic therapies applied ex vivo,[395] and biomarkers of the perfusion process that can predict outcomes of the organ following transplantation beyond histology.[396]

Xenotransplantation has moved from the preclinical to clinical study stage. Xenotransplantation refers to transplantation of tissue or organs from one species to another. Due to natural antibodies against other species acquired during evolution, such a transplant would undergo immediate ABMR (i.e., hyperacute rejection). Multiple models and strategies have evolved over several decades to mitigate this antibody response in the host, as well as ameliorate the innate immune response associated with xenotransplant including complement activation and regulation. Studies in brain dead donors for kidney xenotransplants[397,398] as well as cardiac and renal xenotransplants into living human beings,[399,400] have engendered both enthusiasm and ethical debate about the potential of a genetically modified and personalized organ supply that is an unlimited and renewable resource to treat graft failure.[401,402] A thorough discussion of xenotransplant is beyond the scope of this chapter; however, a better understanding of the immunologic mechanisms and immunosuppressive strategies,[401] as well as the potential for zoonosis or infectious transfer of species-specific pathogens, is required.[403] Whether a true advancement will occur remains to be seen. Caution is noted regarding the long-term survivability of these grafts that, in spite of multiple genetic manipulations in the donor and global posttransplant immunosuppression, appear to still develop ABMR, a barrier to long-term success.[404]

While transplant tolerance has remained elusive as a clinically translatable and practical strategy, there has been continued work including clinical studies examining strategies to induce microchimerism using a number of different donor cell types, as well as augmenting regulatory T cells with some limited successes in select patients.[405] Preclinical strategies using chimeric antigen receptor T cells (CAR Tregs) represent a new strategy for immunosuppressive management,[406] leveraging experiences in cancer and autoimmune trials. A similar approach is under consideration for management of viral infections post transplantation, such as BK and CMV.[407] Whether these will ultimately translate into strategies for patients depends on the relative ease of the approach for both donor and recipient, their overall safety in avoiding overimmunosuppression and graft versus host disease, and whether therapeutic effects can endure without repeated treatments.

Finally, the goal of a precision approach to immunosuppression is challenged by cost of current biomarker assays and insufficient demonstration of clinical utility.[408] Further, effective treatments for DGF, ABMR, and BK virus have not been achieved, in spite of promising preclinical studies. Long-term kidney graft survival may benefit from nonnephrotoxic immunosuppression as noted earlier (refer to earlier section "Costimulatory Signal Blocker"). However, there has been less innovation in posttransplant therapeutics,[409] with barriers identified including the lack of appropriate clinical trial endpoints to facilitate new therapy developments.[410] To this end, there is intense focus on regulation of drug development to support advancements in transplantation. One example is the iBOX, a multicomponent set of clinical parameters that is predictive of late graft failure,[411] which has been accepted as a new secondary endpoint by the Europeans Medicine Agency and is under review at the U.S. Food and Drug Administration.[412]

CONCLUSION

Improvements in short- and long-term kidney allograft survival have been encouraging. This reflects multiple influences including more effective immunosuppression, more frequent use of living donors, and better medical and surgical care. The focus is likely to shift somewhat toward improving other posttransplant outcomes, such as complications of immunosuppression, chronic allograft dysfunction, and morbidity from cardiovascular disease. Availability of adequate numbers of organs for transplantation remains an ongoing problem.

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